



Metal-Organic Frameworks (MOFs)

- from basic research to potential applications

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University of Düsseldorf, Germany

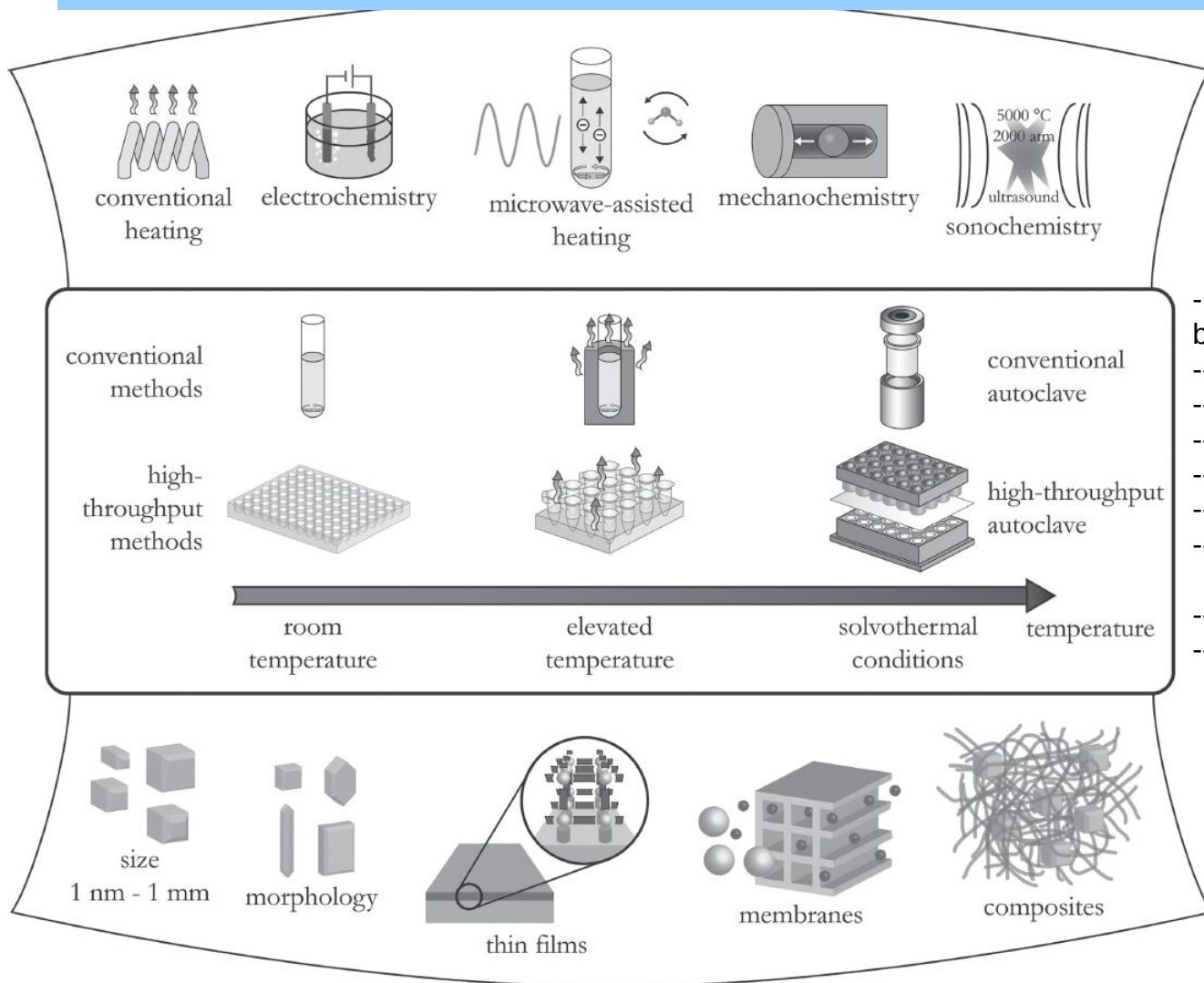
Email: janiak@uni-duesseldorf.de



Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected potential applications

Overview of synthesis methods, possible reaction temperatures, and final reaction products in MOF synthesis



- different synthesis methods that have been applied in the last 20 years.
- room temperature synthesis,
- conventional electric (CE) heating,
- microwave (MW) heating,
- electrochemistry (EC),
- mechanochemistry (MC), and
- ultrasonic (US) methods
- conventional step-by-step methods
- high-throughput methods

MOF synthesis

- is the term "design" justified?

“design”?

– often used term in MOF synthesis and structures
(as in many other fields of chemistry)

-controversial

- a more stringent definition of “design”, implies “to create, fashion, execute, or construct according to plan”
- generally not applicable in MOF synthesis.

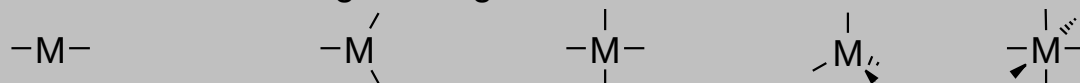
- **at present: knowledge** of possible topologies, the functionality of multitopic organic linker molecules, as well as the understanding of typical metal coordination environments or the formation conditions of typical inorganic building blocks (sometimes termed secondary building units) help one **to understand and direct the synthesis efforts**.

- "design" valid only for isorecticular syntheses

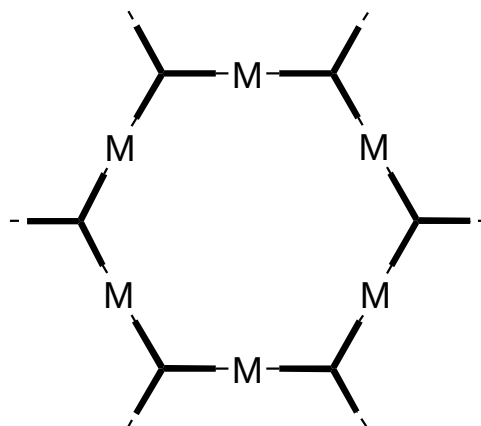
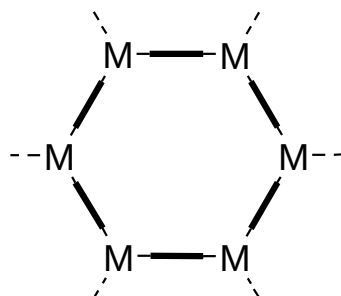
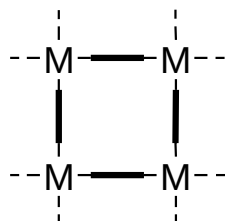
Metal-organic networks:

goal: design through nodal properties of metal and ligands

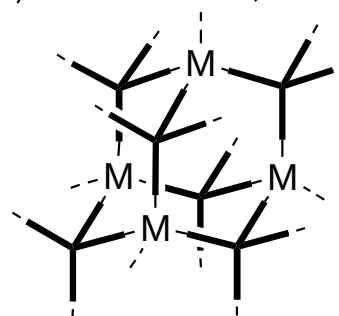
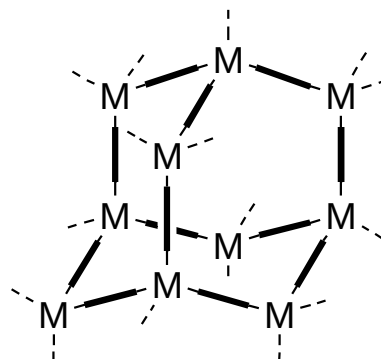
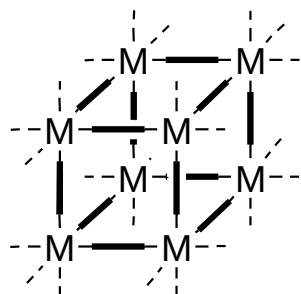
Metal ions or metal-ligand fragments with free coordination sites



1D chain



2D net



3D framework

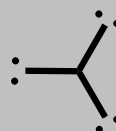
Eddaoudi et al.,
Acc. Chem. Res.
2001, 34, 319

Oxtoby et al.
PNAS **2002**,
99, 4905;

Eddaoudi et al.,
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James,
Chem. Soc. Rev.
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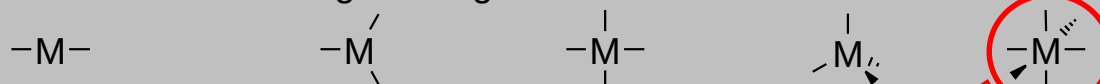
bridging ligands



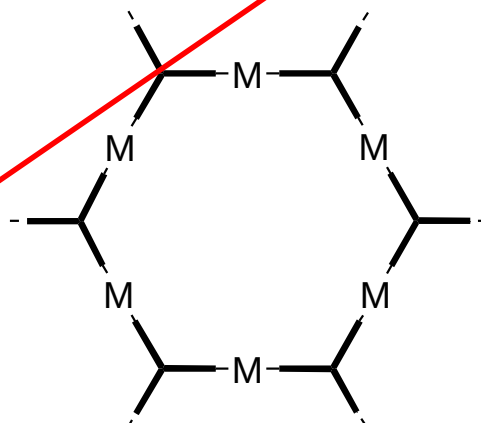
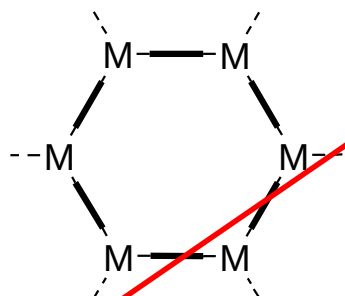
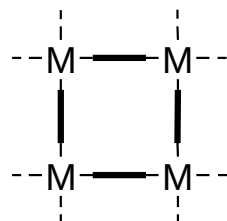
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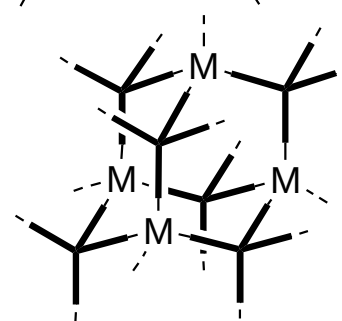
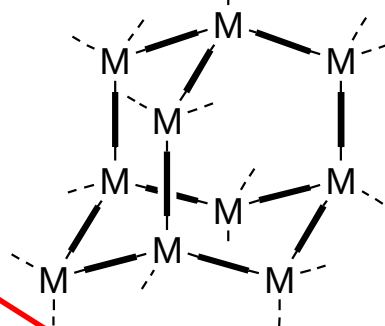
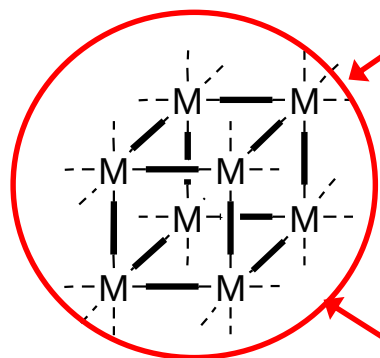
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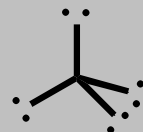
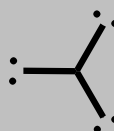


2D net



3D framework

bridging ligands



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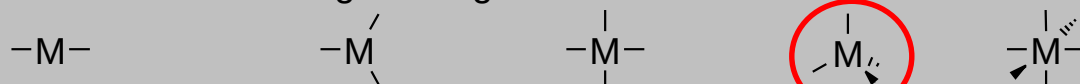
James,
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Janiak,
Dalton Trans. **2003**,
2781 DOI:
10.1039/b305705b.

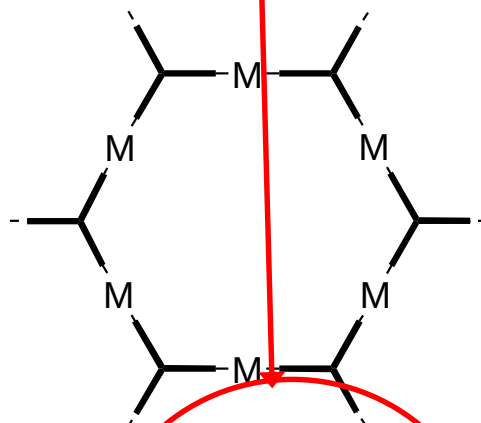
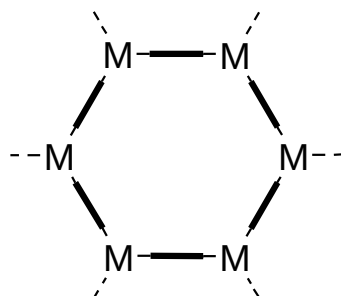
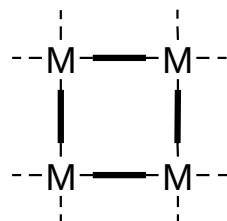
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goal: design through nodal properties of metal and ligands

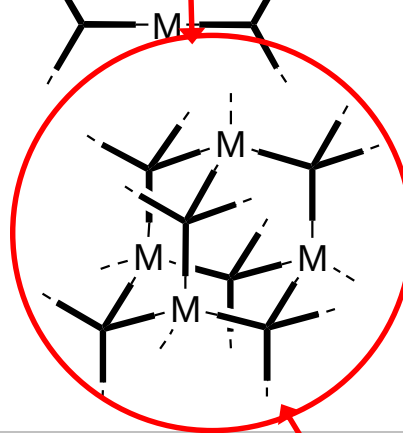
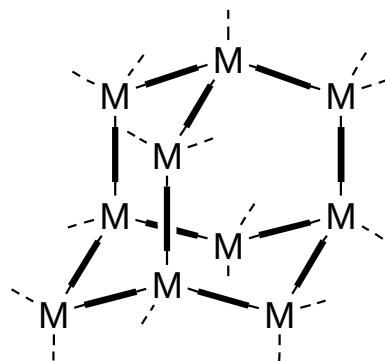
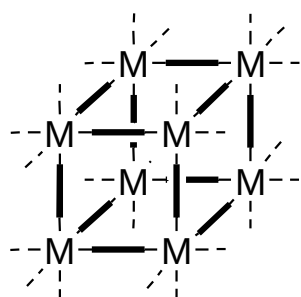
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3D framework

bridging ligands



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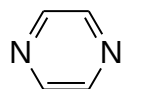
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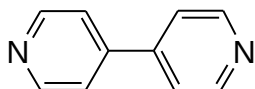
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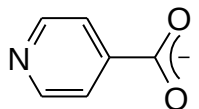
Metal-organic networks: - rigid linkers help in the "design"



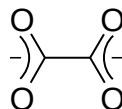
pyrazine



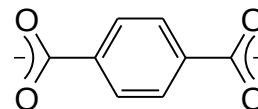
4,4'-bipyridine



isonicotinate



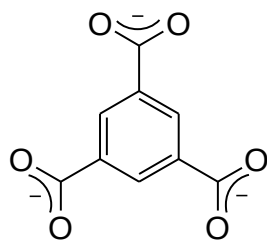
oxalate



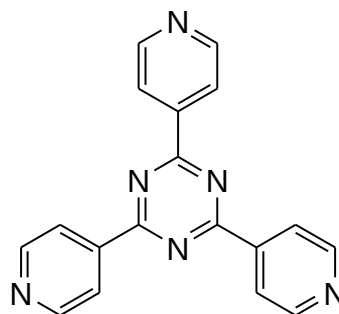
benzene-1,4-dicarboxylate,
terephthalate

linear rigid

(also
linear flexible
possible)

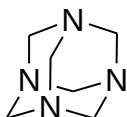


benzene-1,3,5-tricarboxylate,
trimesate

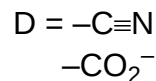
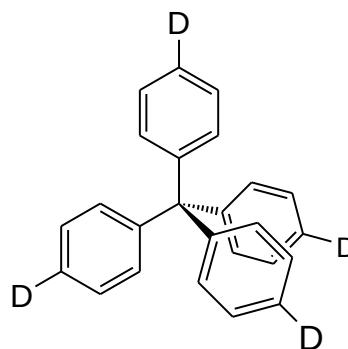


2,4,6-tris(4-pyridyl)-
1,3,5-triazine

trigonal



hexamethylene-
tetramine



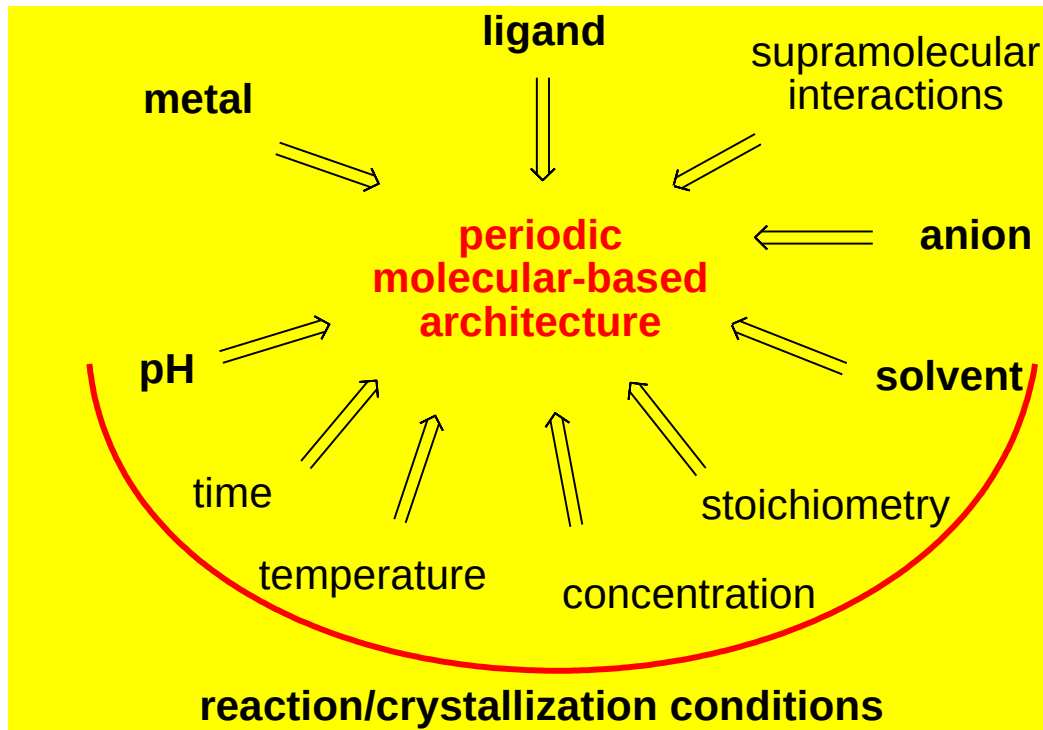
tetra(4-cyanophenyl)methane
tetra(4-carboxyphenyl)methane

tetrahedral

Oxtoby et al., *PNAS* **2002**, 99, 4905; Kitagawa, Masaoka, *Coord. Chem. Rev.* **2003**, 246, 73;
Barnett, Champness, *Coord. Chem. Rev.* **2003**, 246, 145; Zheng et al., *Coord. Chem. Rev.* **2003**, 246, 185;
Ye et al., *Coord. Chem. Rev.* **2005**, 249, 545; Zhang, Chen, *Chem. Commun.* **2006**, 1689

MOF synthesis

control / conditions:



- from the same reaction mixtures can lead to different MOFs.
- may have a strong impact on
 - reaction time,
 - yields, or
 - particle size and
 - morphology.

MOFs:

- effect of solvent (changes)

- solvent is important as **template / porogen**
- solvent is initially **contained in pores** (before activation)
- solvent is **terminal ligand** for otherwise coordinatively unsaturated metal sites (CUS)
- few systematic studies on solvent effects:

system Fe³⁺/H₂bdc-NH₂/solvent:

- in **acetonitrile** as well as in methanol, at low temperatures, only **Fe-MIL-88B-NH₂**.
- with **DMF** **Fe-MIL-88B-NH₂** and **Fe-MIL-101-NH₂**, were isolated,
- the reaction in **water** led to **Fe-MIL-53-NH₂** in most of the cases.

system Al³⁺/H₂bdc-NH₂/solvent:

- **aluminum nitrate** using **DMF**, **methanol**, and **water** led to **Al-MIL-53-NH₂**;
- **aluminum chloride** in **DMF** gave **both Al-MIL-53-NH₂** and **Al-MIL-101-NH₂**.
- reactions performed in **methanol** yielded **CAU-1**.

MOF synthesis

- **complex relationships**

- hence, only for selected examples MOFs of a same structure could be obtained
- MOFs of a same structure are called **isorecticular**
- the discovery of the majority of MOFs is still **exploratory**
- **a lot of** synthetic **attempts** are usually mandatory.
- the **low success rate** (in terms of new crystalline MOF) is perhaps 10% (or less)
- to speed up the discovery of MOFs, **high-throughput methods**, which were first developed in pharmaceutical research, may be used (cf. N. Stock, Chem. Rev. 2012)

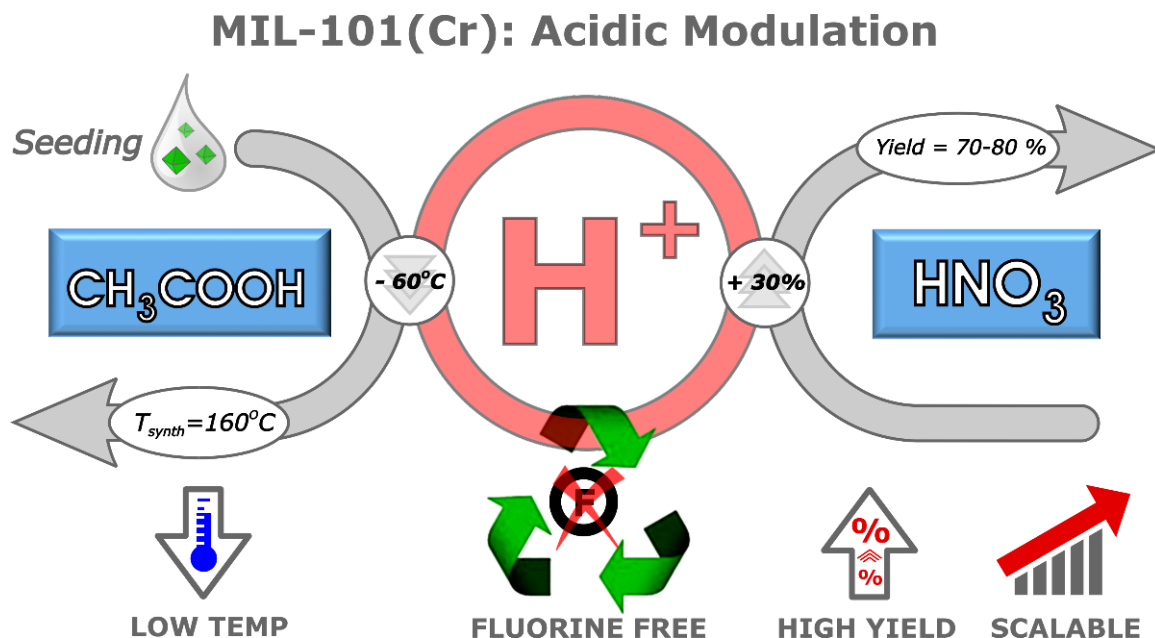
MOF synthesis

- objectives – besides just synthesis

- to obtain – at least for the first time synthesis – (single) crystals for X-ray diffraction
- high yield and larger bulk quantities
- lower temperature / ambient pressure to avoid hydrothermal conditions
- continuous synthesis
- faster reactions, decrease synthesis times
- to simplify / speed up activation
- to use less solvent, also for activation
- fluorine-free synthesis

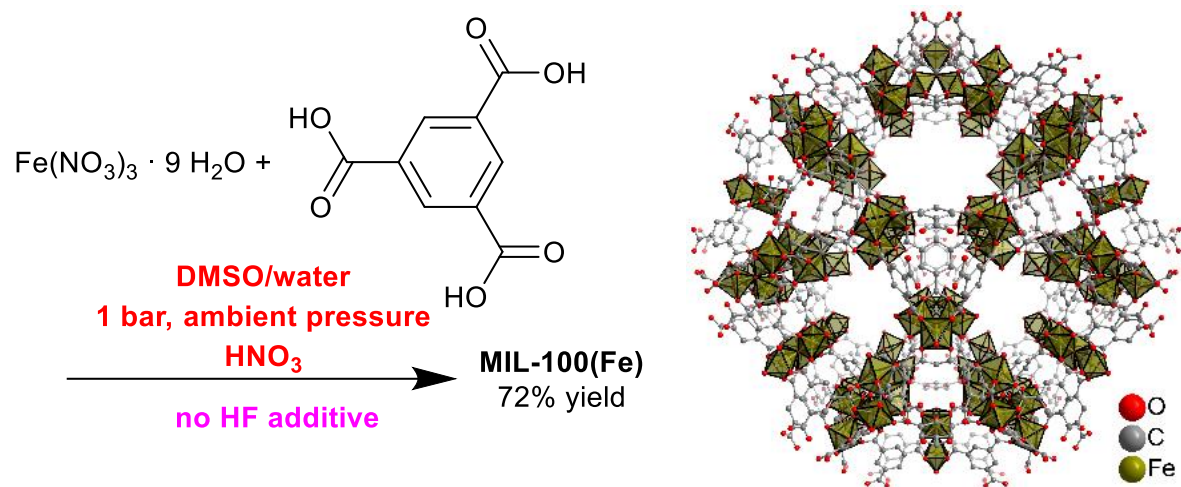
Example MIL-101(Cr) synthesis

- high-yield, fluoride-free and large-scale

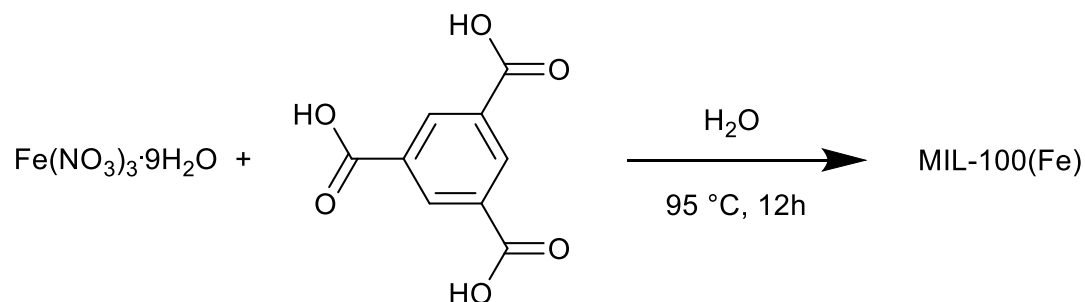


Dalton Trans. **2015**, 44, 16791-16801. DOI: 10.1039/C5DT02625C

Example MIL-100(Fe) - ambient pressure synthesis



Dalton Trans. **2016**, 45, 8637-8644. DOI: 10.1039/C6DT01179A



F. Zhang, J. Shi, Y. Jin, Y. Fu, Y. Zhong, W. Zhu, *Chem. Eng. J.*, **2015**, 259, 183-190.



MOF synthesis

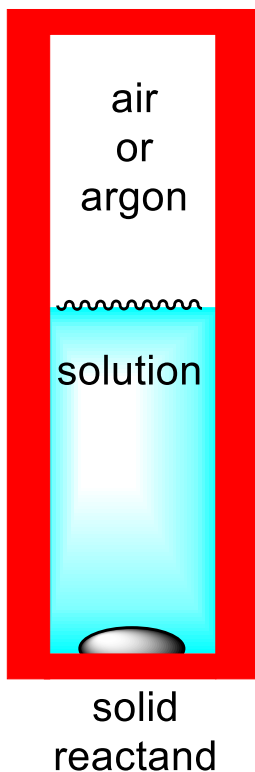
- objectives – besides just synthesis

- (crystal) size, shape, morphology
- high surface area for given MOF product
- thin film preparation
- membranes,
- composite materials

MOF synthesis

- hydrothermal synthesis in autoclave

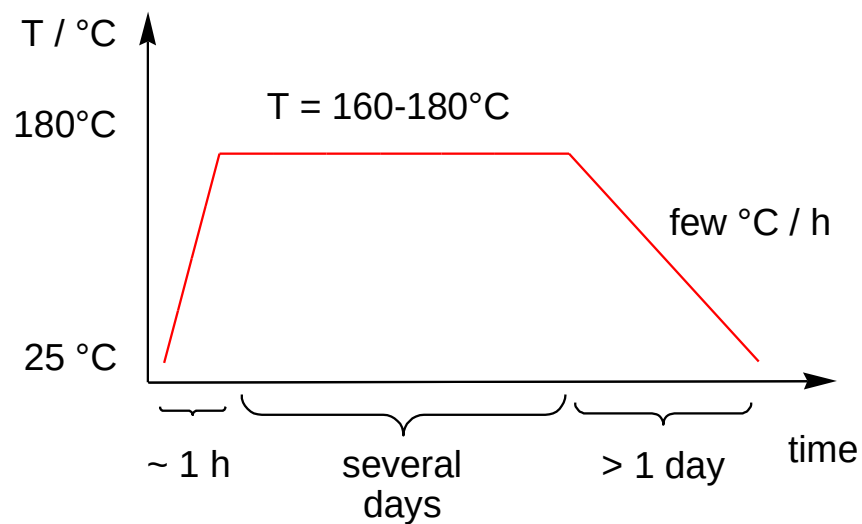
autogenous pressure above the boiling point of the solvent.



Teflon liner
in steel autoclave



Temperature profile – programmable heating oven



MOF – hydrothermal synthesis equipment

programmable heating ovens for hydrothermal synthesis:



problems:

- expensive equipment
- only bulk synthesis
- upscaling limited
- long reaction times

steel autoclaves



3 liter
autoclave
reactor



used in *Dalton Trans.* **2015**, 44, 16791-16801.
DOI: 10.1039/C5DT02625C

MOF synthesis

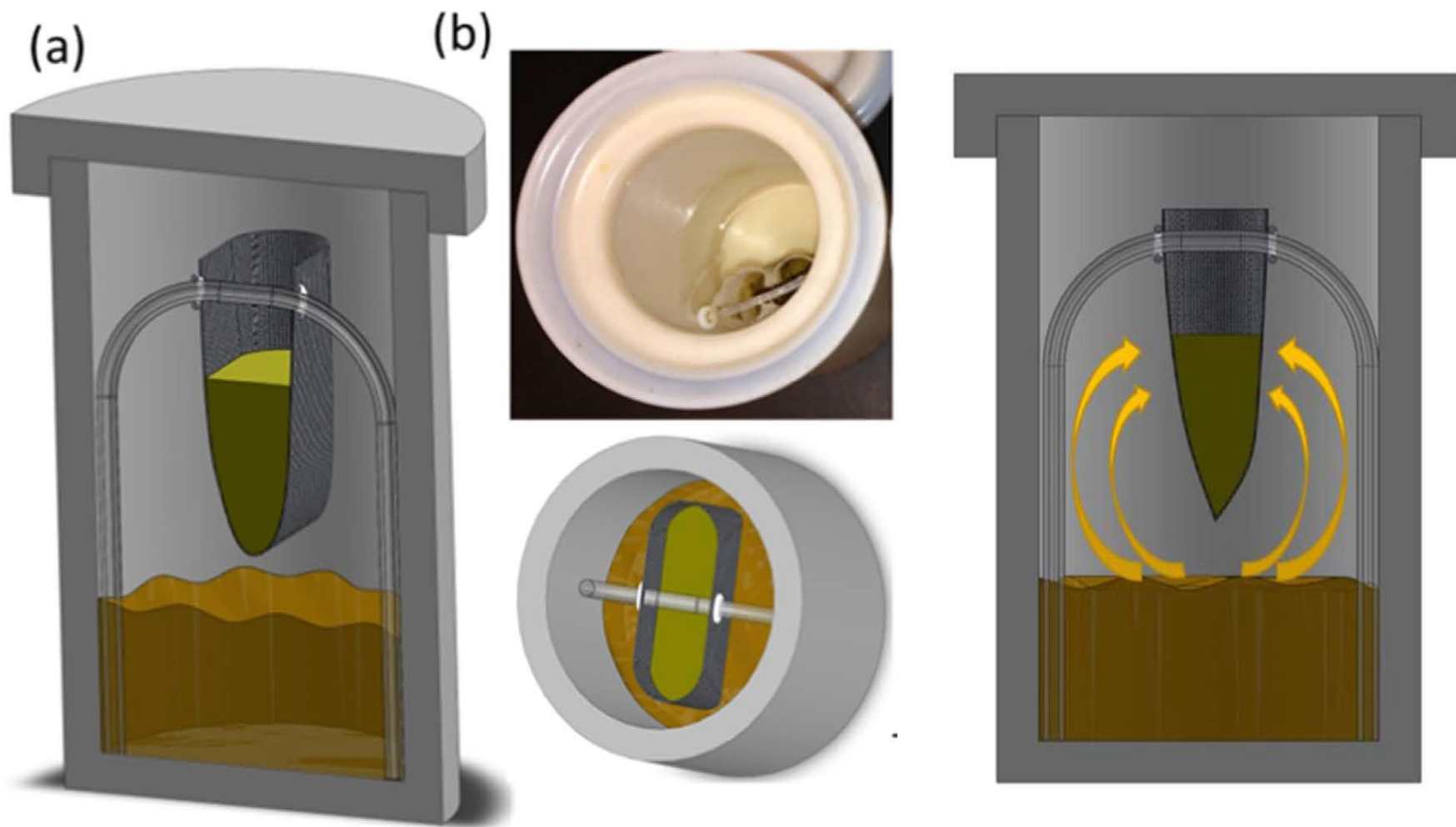
- reaction scale-up

The efforts in recent years have resulted in the scale-up of well known MOFs in good space time yields, such as MOF-5, Al-MIL-53, HKUST-1, ZIF-8, and Fe-MIL-100 and some products are also commercially available

Table 1. Summary of MOFs That Have Been Synthesized in Larger Amounts

MOF	comment
MOF-2	organic solvent, open vessel, 87% yield (Zn)
MOF-5	organic solvent, open vessel, 91% yield (Zn)
IRMOF-8	organic solvent, open vessel, 15% yield (Zn)
Al-MIL-53	commercially available as Basolite A100
HKUST-1	solvothermal batch synthesis commercially available as Basolite C 300 organic solvent, electrochemical synthesis continuous synthesis; microwave-assisted heating
Fe-BTC	commercially available as Basolite F300
ZIF-8	commercially available as Basolite Z1200
[Mg(O ₂ CH) ₂]	commercially available as Basosiv M050
Fe-MIL-100	hydrothermal reaction, 10 L reactor hydrothermal reaction, 100 L reactor

MOF – hydrothermal synthesis by dry-gel conversion (DGC)



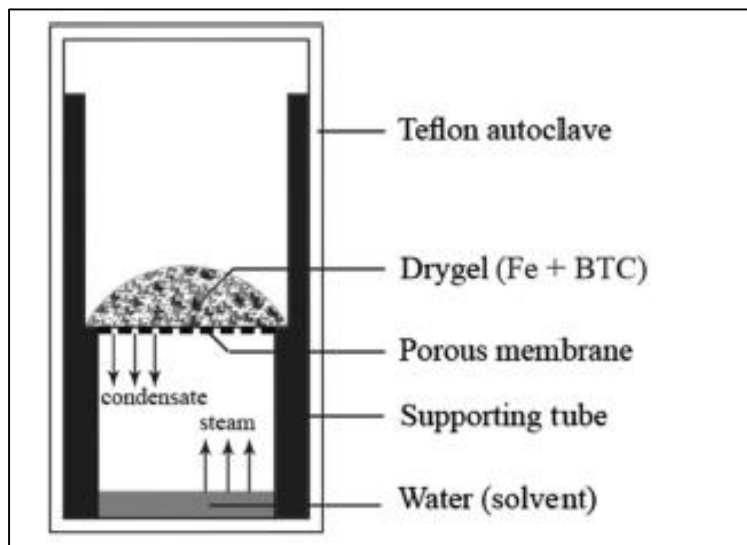
MOF – hydrothermal synthesis by dry-gel conversion (DGC)

Advantages

- Recovery of solvents
- Shorter reaction time
- Higher yield

Literature known MOFs synthesized via DGC

- MIL-100(Fe)
- MIL-101(Cr)
- MIL-125(Ti)
- MOF-74
- ZIF-8
- ZIF-67

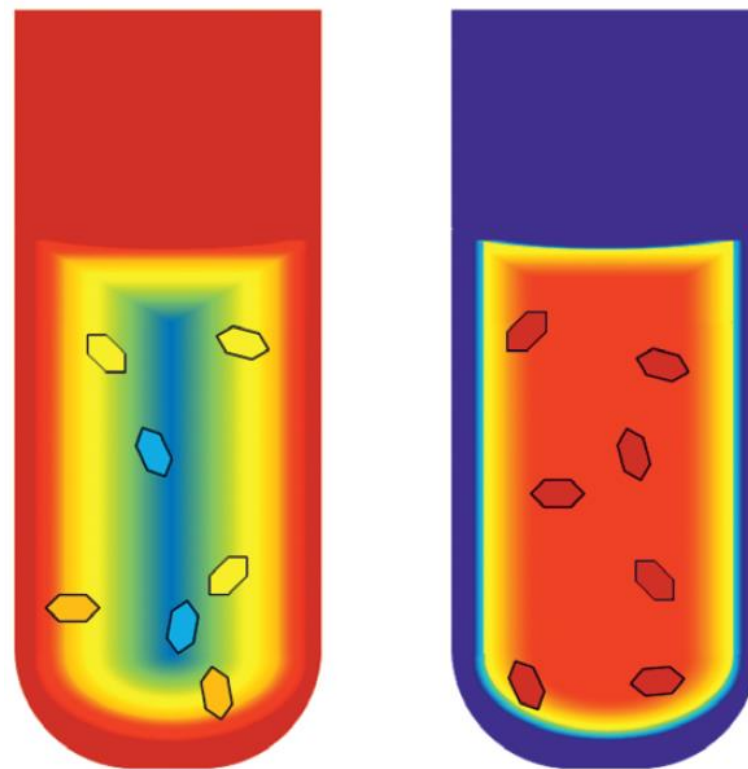


DGC of MIL-100(Fe)

MOF synthesis

- microwave-assisted synthesis

- acceleration reaction (times < 1 h)
- the formation of nanoscale products
- to improve product purity and



temperature profile

conventional heating

μ-wave heating

M. F. Groh, M. Heise, M. Kaiser, M. Ruck,
Nachr. Chemie **2013**, 61, 26-29.

MOF synthesis

- microwave-assisted synthesis

partial Table taken from literature

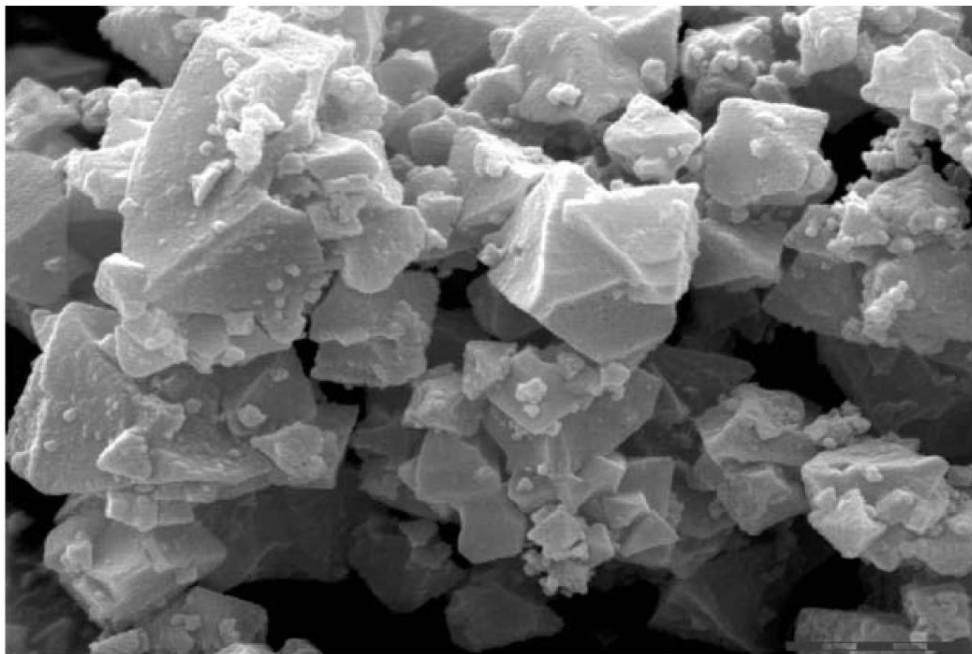
Table 5. Summary of Studies Dealing with the Microwave-Assisted Synthesis of Porous MOFs

MOF	comment
Metal(III)—Carboxylate-Based MOFs	
Cr—MIL-100	first MW synthesis of MOFs
Fe—MIL-100	nanoscale particles; imaging and drug delivery
Cr—MIL-101	nanoscale particles; benzene sorption
	nanoscale particles
Fe—MIL-101	nanoscale particles
Fe—MIL-101—NH ₂	nanoscale particles; imaging and drug delivery
	nanoscale particles; imaging and drug delivery
Fe—MIL-88A	CE, MW and US synthesis; nanoscale particles
Fe—MIL-53	CE, MW and US synthesis; ex-situ crystallization study
CAU-1	CE and MW synthesis; in situ crystallization study
V—dicarboxylates	isorecticular frameworks with MIL-47 topology
	growth on polyacrylonitrile
[Ce ₂ (PYDC) ₂ (HPYDC) (H ₂ O) ₂]Cl	lanthanide-based MOF

MOF synthesis

- electrochemical synthesis

industrial Zn- and Cu-MOF synthesis by BASF SE



20000 : 1

1 μm

SEM-picture of Cu-MOF crystals from electrochemical preparation (scale bar: 1 μm).

Synthesis cell for electrochemical preparation of MOFs with Cu-plates as electrode material (laboratory scale).



MOF synthesis

- HKUST-1, Cu-btc electrochemical synthesis

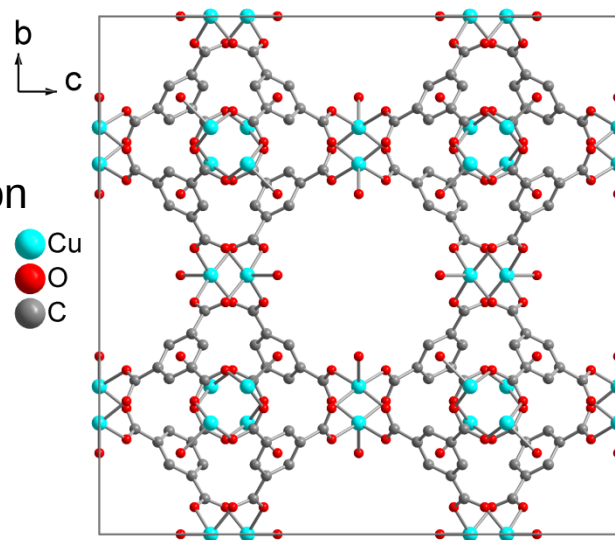
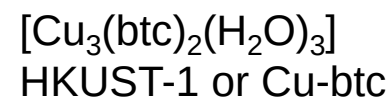
HKUST-1, Cu-btc:

At BASF this Cu-MOF was for the first time synthesized using an electrochemical route.

Bulk copper plates, thickness 5 mm, are arranged as the anodes in an electrochemical cell with the carboxylate linker, viz. 1,3,5-benzenetricarboxylic acid, dissolved in methanol as solvent and a copper cathode.

Experimental:

During a period of 150 min at a voltage of 12–19 V and a current of 1.3 A, a greenish blue precipitate was formed. After separation by filtration and drying at 120 °C overnight pure Cu-MOF was obtained. The surface area determination yielded 1649 m² g⁻¹ and after activation at 250 °C finally 1820 m² g⁻¹ giving a dark blue colored solid of octahedral crystals from 0.5 to 5 mm size.



MOF synthesis

- mechanochemical synthesis

Mechanical force can induce many physical phenomena (mechanophysics) as well as chemical reactions. In mechanochemical synthesis, the mechanical breakage of intramolecular bonds followed by a chemical transformation takes place. Mechanochemistry has a long history in synthetic chemistry and it has recently been employed in multicomponent (ternary and higher) reactions to form pharmaceutically active cocrystals and in inorganic solid-state chemistry, organic synthesis, and polymer science, just to name a few. Its use for the synthesis of porous MOFs was first reported in 2006.

MOF synthesis

- mechanochemical synthesis

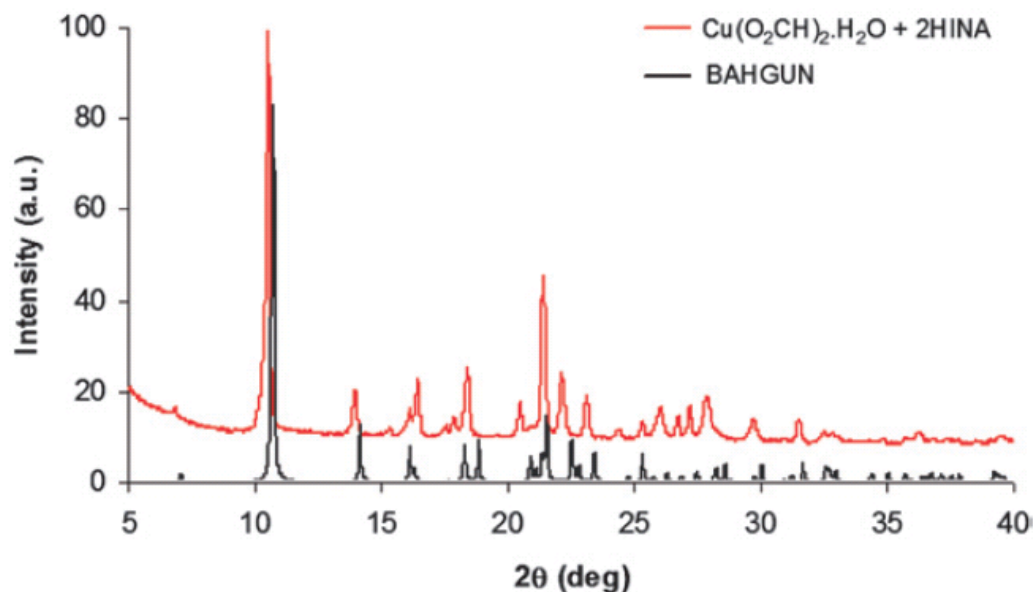
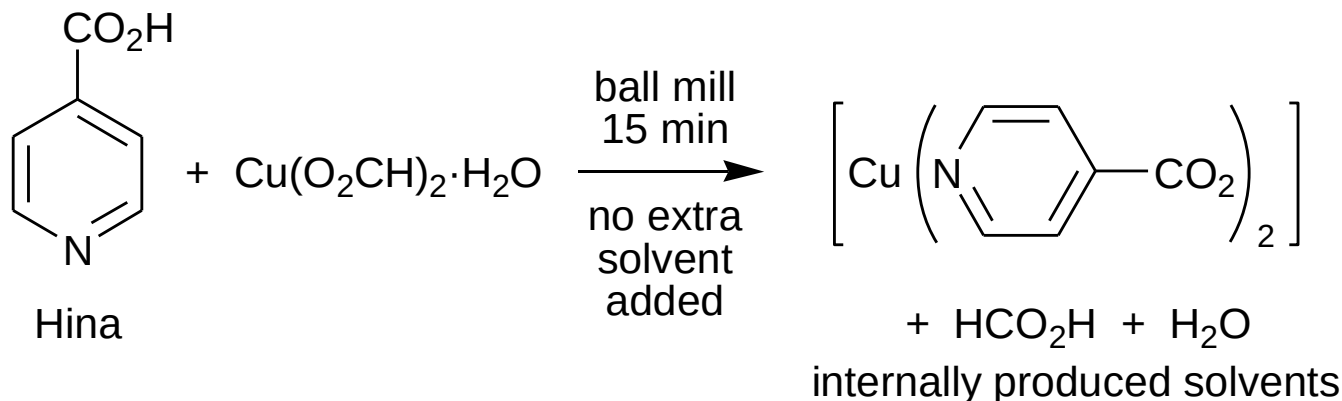
MOFs can also be prepared using mechanochemistry, that is, the grinding of two or more solids by using, for example, a mechanical ball mill and thereby avoid the use of solvents.



Ball mills which can generate microcrystalline products within minutes.

MOF synthesis

- mechanochemical synthesis



Synthesis of $[\text{Cu}(\text{ina})_2]$ (top) and comparison of the XRPD pattern of the product and the simulated pattern for $[\text{Cu}(\text{ina})_2] \cdot 2\text{H}_2\text{O}$ (CSD-Refcode BAHGUN) based on single-crystal data.

also
 ZIF-8
 HKUST-1
 prepared by
 grinding

A. Pichon, A. Lazuen-Garay, S. L. James, *CrystEngComm*, **2006**, 8, 211.

A. Pichon, S. L. James, *CrystEngComm*, **2008**, 10, 1839-1847.

MOF synthesis

- by twin- or single-screw extrusion (TSE, SSE)

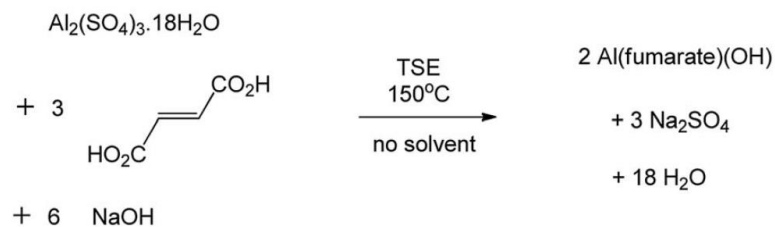
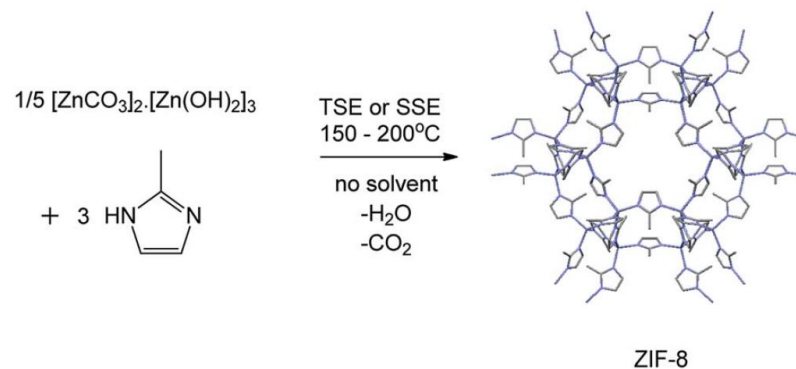
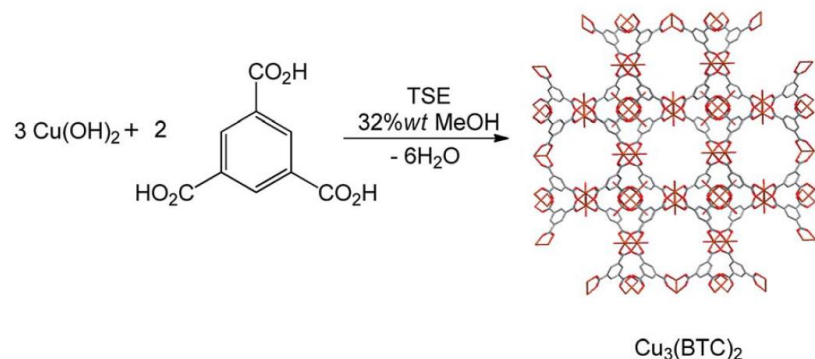


- matching PXRD, high space-time yield

MOF	TSE surface area [m ² g ⁻¹]	Solvothermal surface area ^[61] [m ² g ⁻¹]
HKUST-1	1738	1500–2100
ZIF-8	1603	1300–1800
Al(fumarate)(OH)	1010	1020

D. Crawford, J. Casaban, R. Haydon, N. Giri, T. McNally, S. L. James, *Chem. Sci.*, **2015**, 6, 1645–1649.

D. Crawford, J. Casaban, *Adv. Mater.* **2016**, 28, 5747–5754.



MOF synthesis

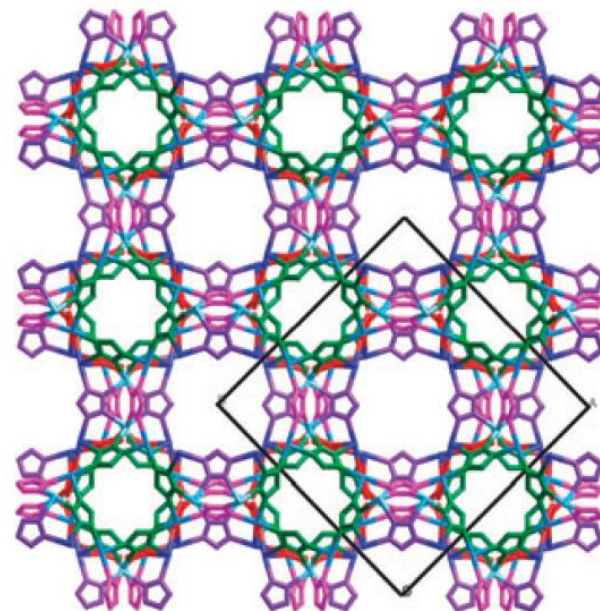
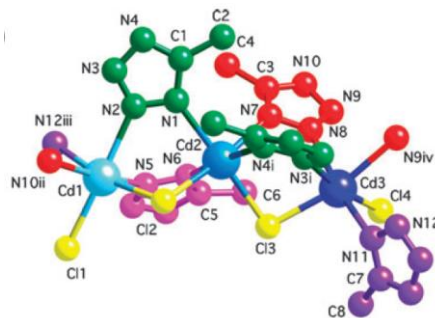
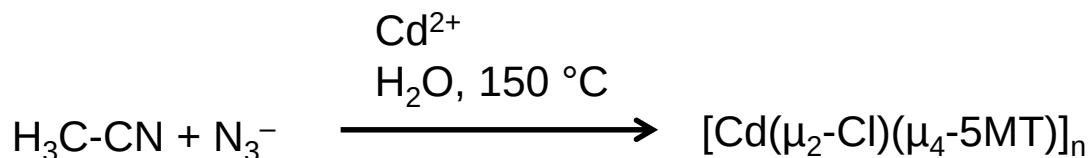
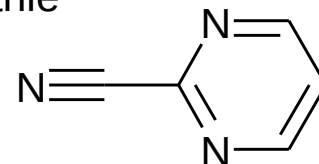
- sonochemical synthesis

Table 8. Summary of Studies Dealing with the Sonochemical Synthesis of Porous MOFs

MOF	comment
Ditopic Linker	
MOF-5	variation of reaction time and temperature as well as power level
IRMOF-9 and -10	3 ² statistical design used for synthesis optimization; consecutive application of US and MW irradiation
Fe—MIL-53	control of catenation and its effect on CO ₂ adsorption; role of power level in US synthesis
Fe—MIL-88A	crystallization kinetics using CE, MW, and US methods; combination of MW and CE heating for nucleation/crystal growth; temperature control; variation of power level
	nanoparticles; flexible porous MOF; CE, MW, and US synthesis
Tritopic Linker	
[Zn ₃ (BTC) ₂]	synthesis of nanoparticles; selective sensing of organoamines
HKUST-1	time-dependent formation in DMF/ethanol/water mixture
	rapid room temperature synthesis; amount of DMF determines nature and morphology of reaction product
[Ln(BTC)(H ₂ O)] (Ln = Ce, Tb, Y)	competitive formation with [Cu ₂ (BTC)(OH)(H ₂ O)] using six synthesis methods
MOF-177	short reaction time; particle size; luminescence
	comparison among CE, MW, and US syntheses; short reaction time; high yield;
	small particles; improved sorption properties
PCN-6 and -6'	control of catenation and its effect on CO ₂ adsorption; role of power level in US synthesis

MOFs: - in situ linker synthesis

- Hydrolysis of esters and nitriles to the corresponding carboxylates
- ester trimethyl-1,3,5-benzenetricarboxylate hydrolyzed to BTC^{3-} for M-MIL-96 ($\text{M} = \text{Al}, \text{Ga}, \text{In}$) and MIL-100(Al), MOFs constructed from Y^{3+} ions and BTC^{3-} linkers.
- rho-type zeolitic MOF $[\text{Cd}(\text{PYMC})]$ obtained by in situ hydrolysis of the nitrile 2-pyrimidininenitrile.

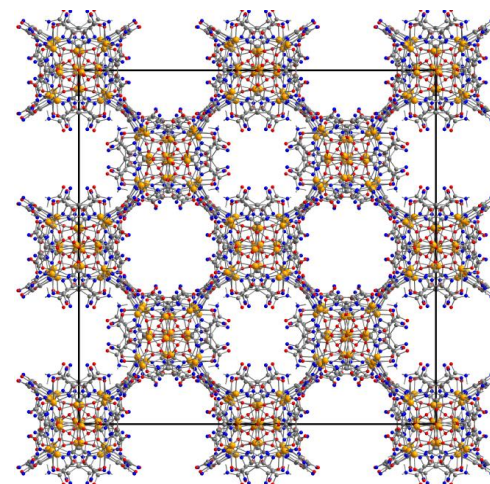
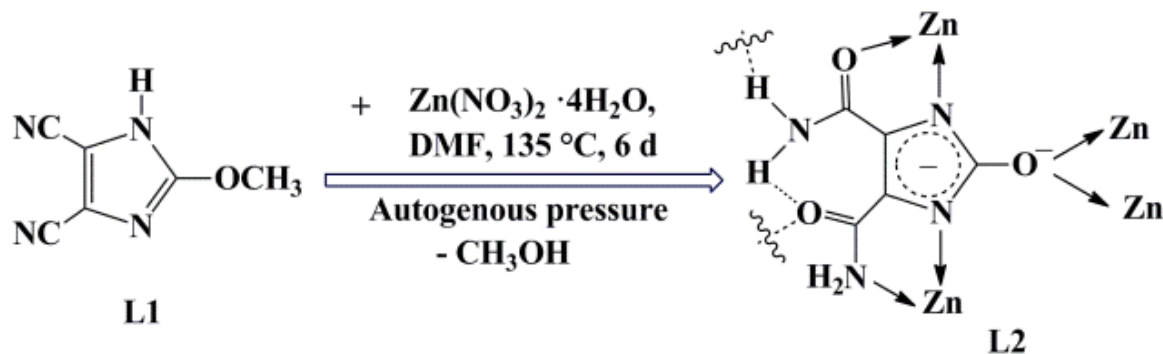
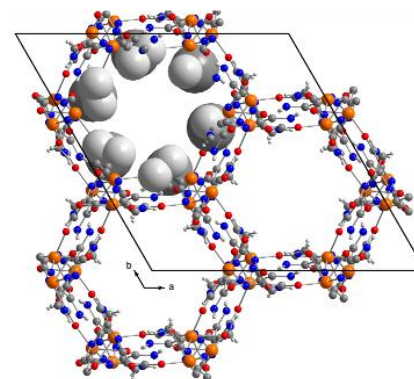
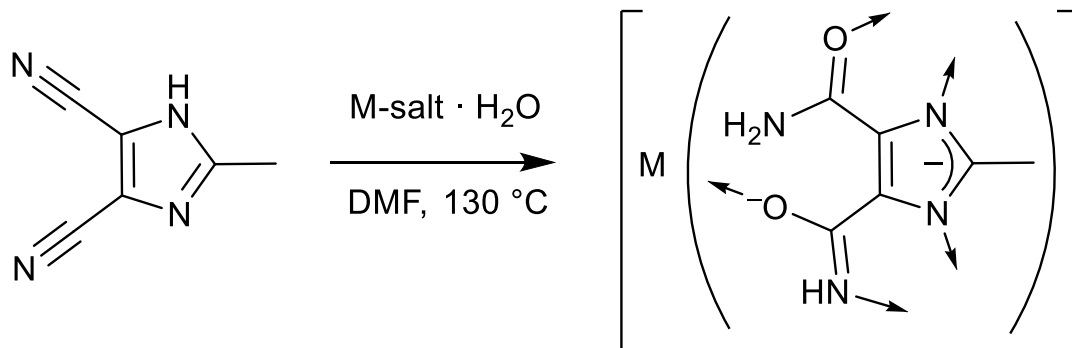


N. Stock, S. Biswas, *Chem. Rev.* **2012**, *112*, 933–969.

Qiu, Y. C.; Deng, H.; Mou, J. X.; Yang, S. H.; Zeller, M.; Batten, S. R.; Wu, H. H.; Li, J. *Chem. Commun.* **2009**, 5415–5417.

MOFs: - in situ linker synthesis

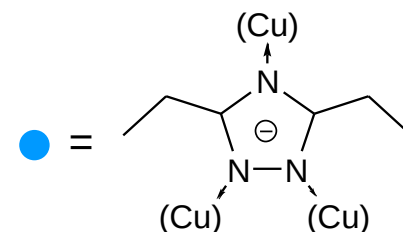
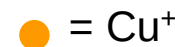
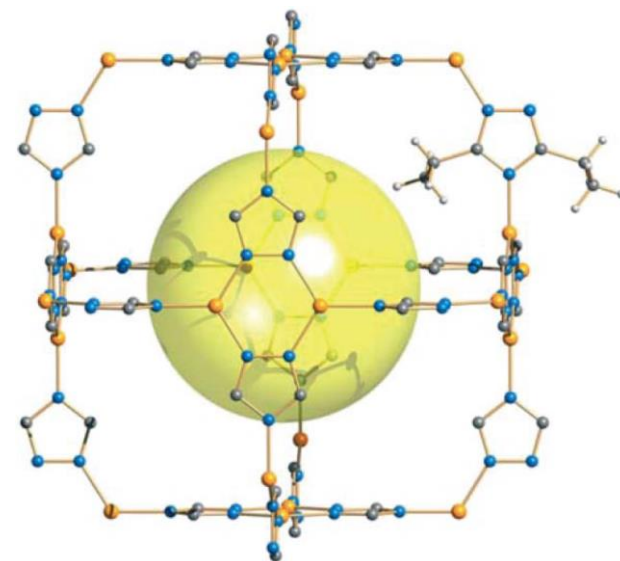
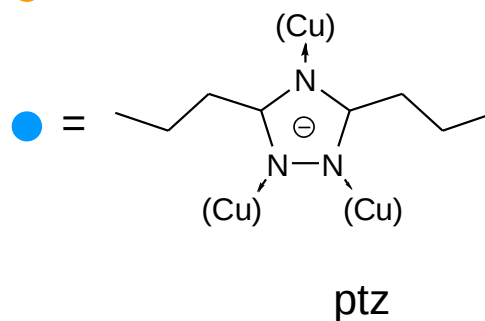
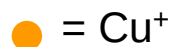
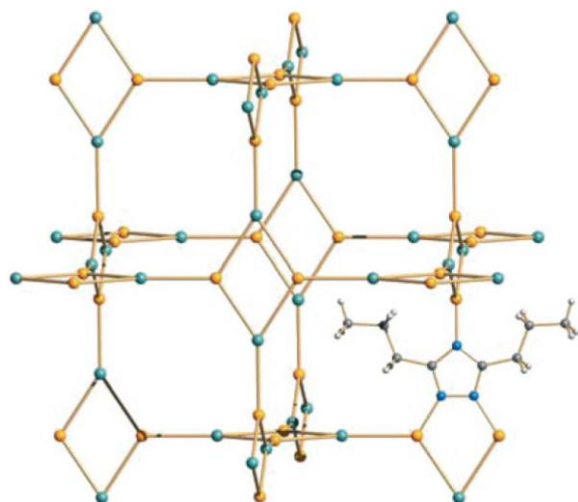
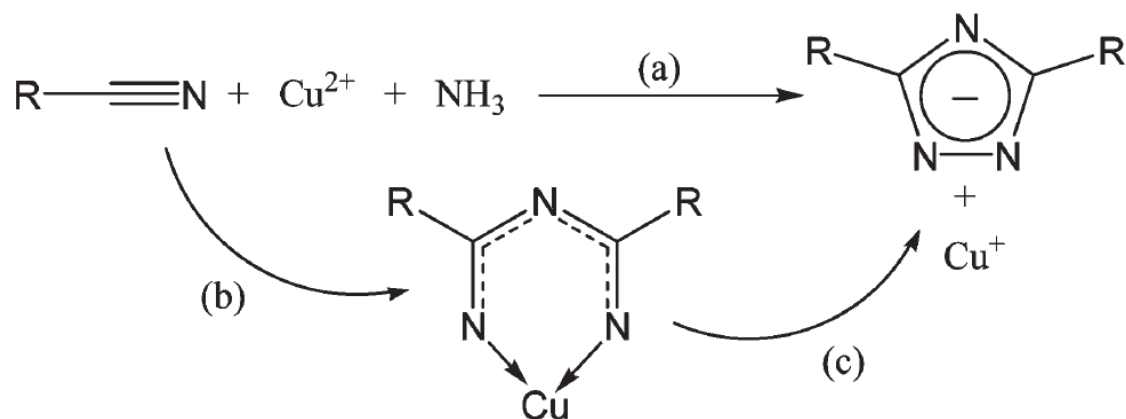
- the imidazolate 4-amide-5-imidate linker, generated in situ by partial hydrolysis of 4,5-dicyano-2-methyl- or methoxy-imidazole



J. Am. Chem. Soc. **2014**, 136, 44-47. DOI: 10.1021/ja410595q
CrystEngComm **2014**, 16, 29-42. DOI: 10.1039/C3CE42040J
CrystEngComm **2013**, 15, 9394-9399. DOI: 10.1039/C3CE41632A
Chem. Commun. **2013**, 49, 7567-7570. DOI: 10.1039/C3CC42156B

Prof. H.J. Holdt

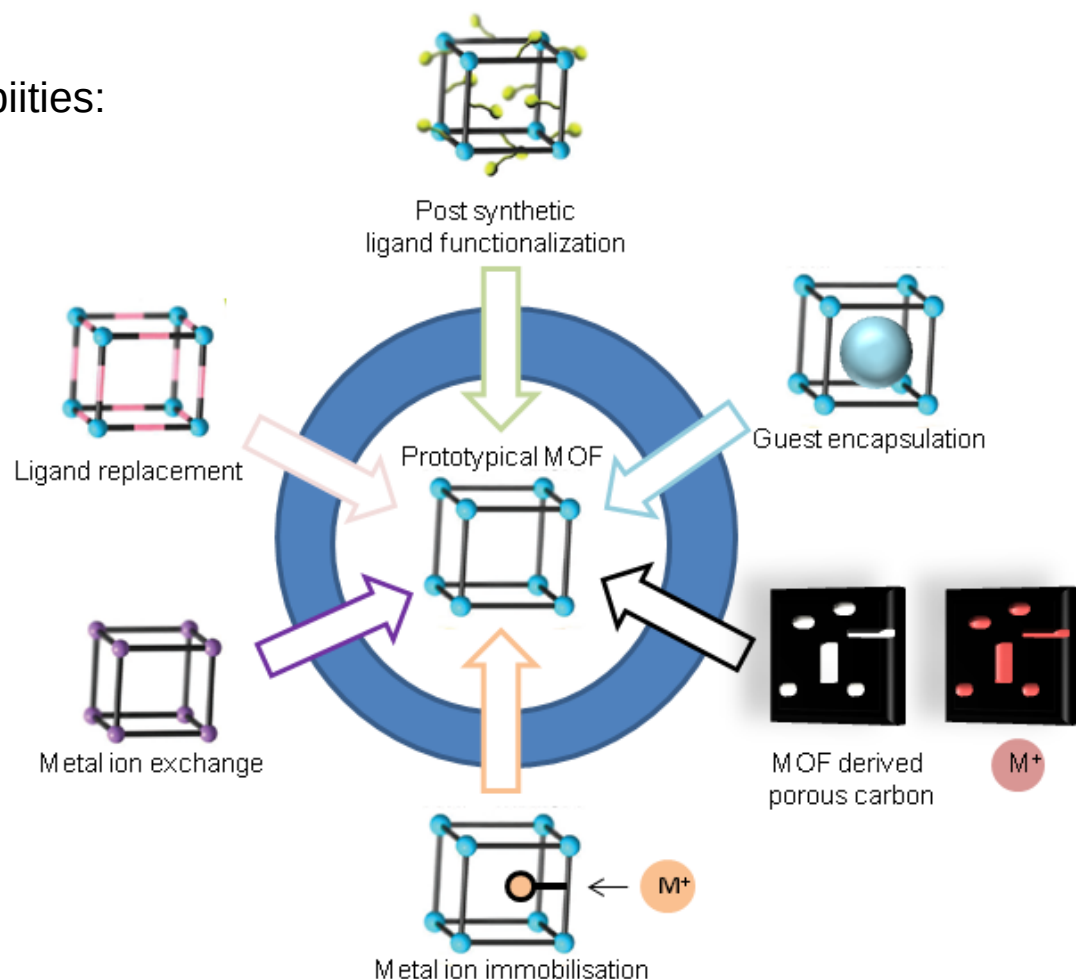
MOFs: - in situ linker synthesis



J.-P. Zhang, X.M. Chen, *Chem. Commun.* (Review) **2006**, 1689.
also Barea, Sironi et al., *Dalton Trans.* **2008**, 1825,

MOF - postsynthetic modification, PSM

principle possibilities:



K. K. Tanabe, S. M. Cohen, *Chem. Soc. Rev.* **2011**, 40, 498–519.

S. M. Cohen, *Chem. Rev.* **2012**, 112, 970–1000.

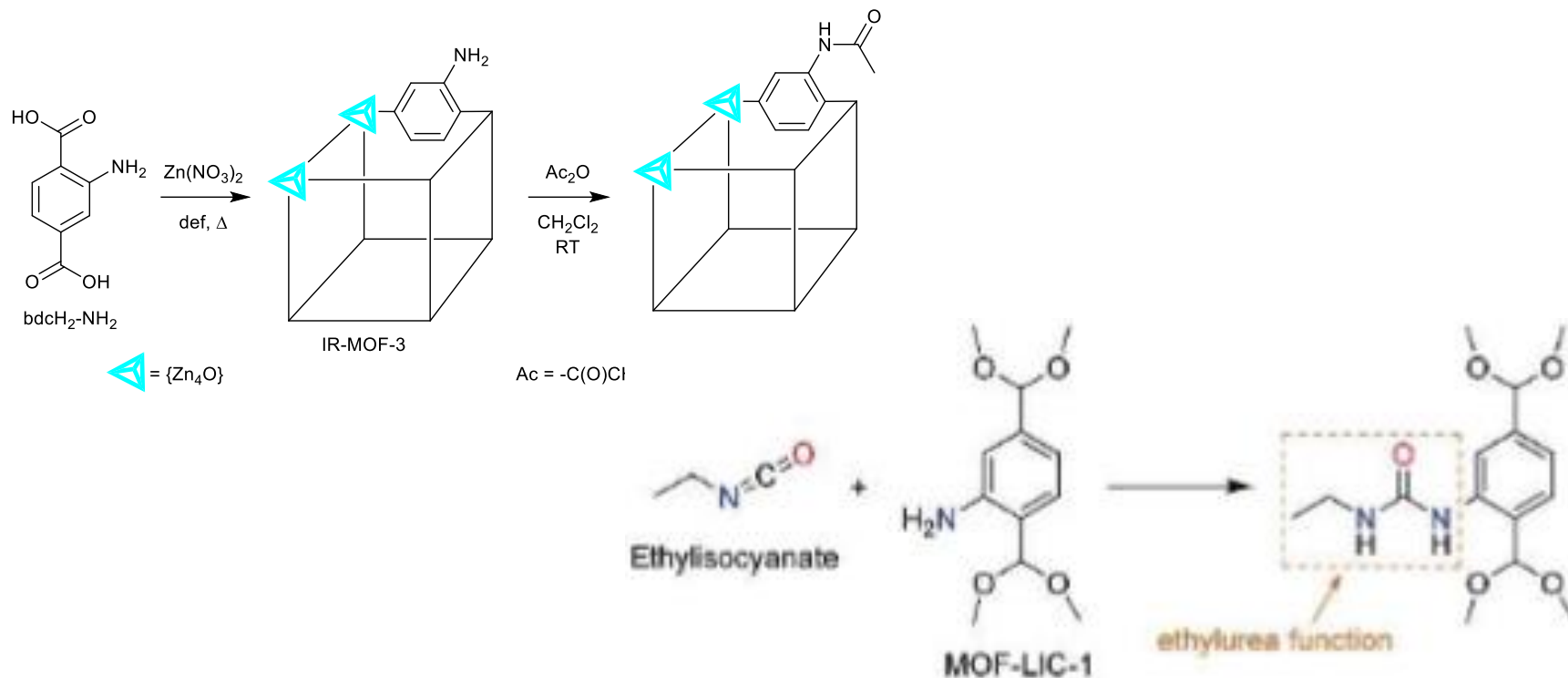
Q. Wang, J. Bai, Z. Lu, Y. Pan, X. You, *Chem. Commun.* **2016**, 52, 443–452.

J. D. Evans, C. J. Sumby, C. J. Doonan, *Chem. Soc. Rev.* **2014**, 43, 5933–5951.

MOF - postsynthetic modification, PSM

- ligand functionalization = covalent postsynthetic modification

- existing stable MOFs are modified after their synthesis
- linker must have functional reactive group
- best-investigated linker for PSM is 2-aminoterephthalate
- large range of synthetic possibilities

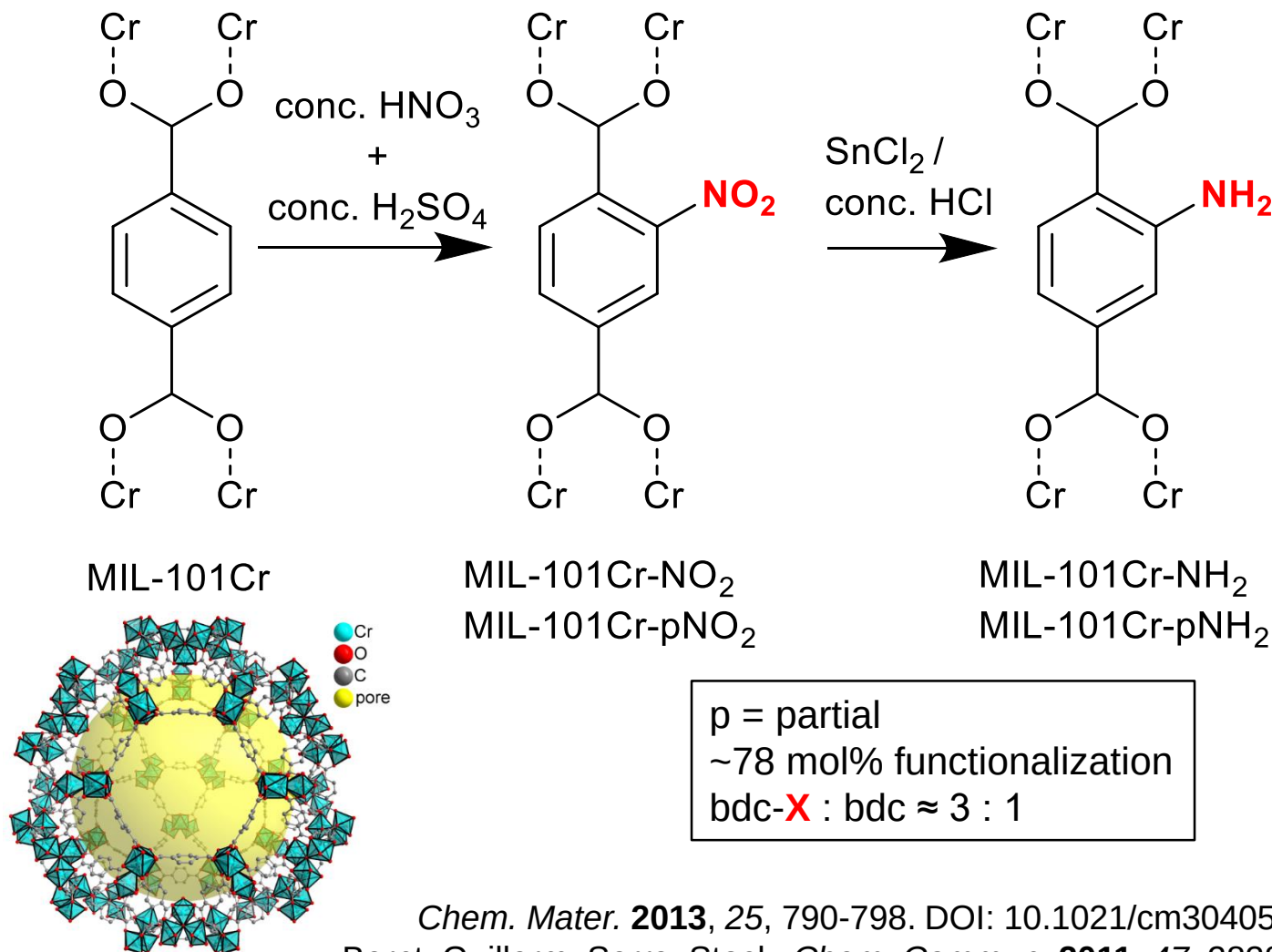


Z. Wang, S. M. Cohen, *Chem. Soc. Rev.*, **2009**, 38, 1315-1329.

J. S. Costa, P. Gamez, C. A. Black, O. Roubeau, S. J. Teat, J. Reedijk, *Eur. J. Inorg. Chem.*, **2008**, 1551.

MOF - postsynthetic modification, PSM - covalent postsynthetic modification

- introducing functionality:



Chem. Mater. **2013**, 25, 790-798. DOI: 10.1021/cm304055k
Bernt, Guillerm, Serre, Stock, *Chem. Commun.* **2011**, 47, 2838–2840.

MOF - postsynthetic modification, PSM - covalent postsynthetic modification

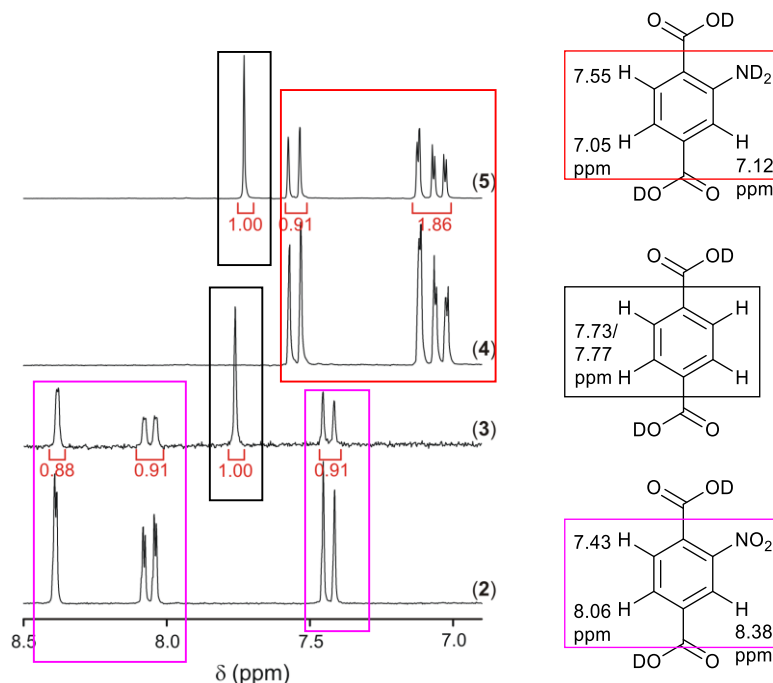
proof and quantification by "digestion" (= destructive dissolution)
for NMR analysis:

MIL-101(Cr)-pNH₂

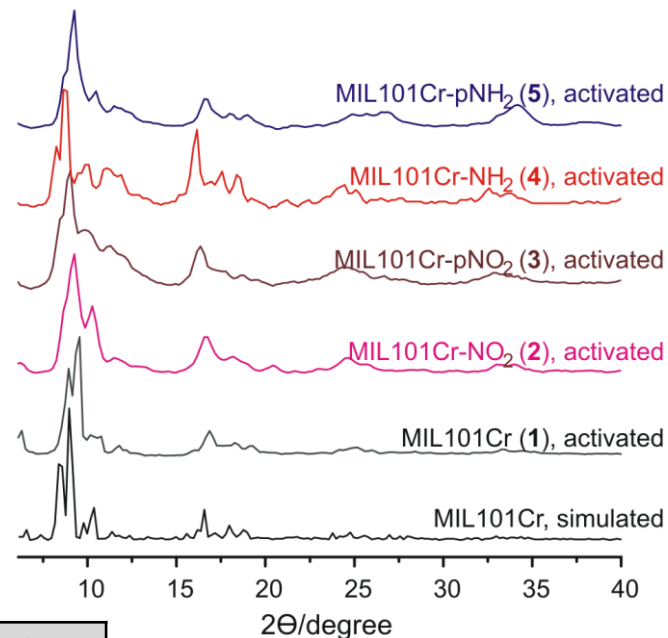
MIL-101(Cr)-NH₂

MIL-101(Cr)-pNO₂

MIL-101(Cr)-NO₂

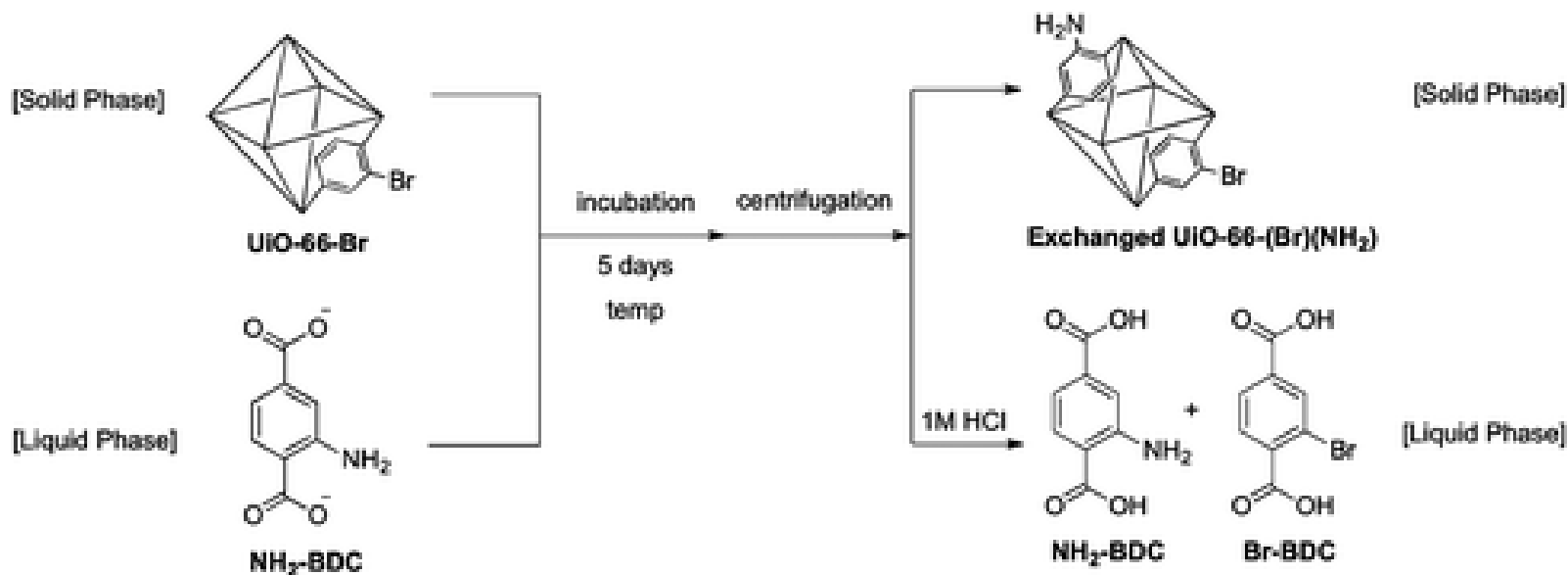


structure retention by PXRD:



Compounds	Pore volume (cm ³ /g)	BET surface area (m ² /g)
MIL-101Cr (1)	1.63	2845
MIL-101Cr-NO ₂ (2)	0.70	1245
MIL-101Cr-pNO ₂ (3)	1.11	2195
MIL-101Cr-NH ₂ (4)	1.6	2700
MIL-101Cr-pNH ₂ (5)	1.44	2495

MOF- postsynthetic modification - solvent-assisted linker exchange (SALE)

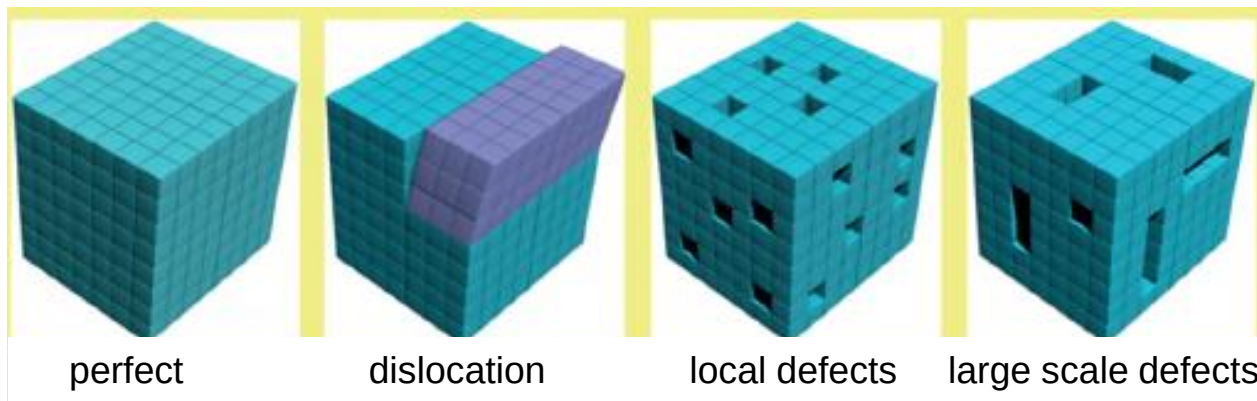


- up to 75% of linker exchanged

MOF

- defect engineering

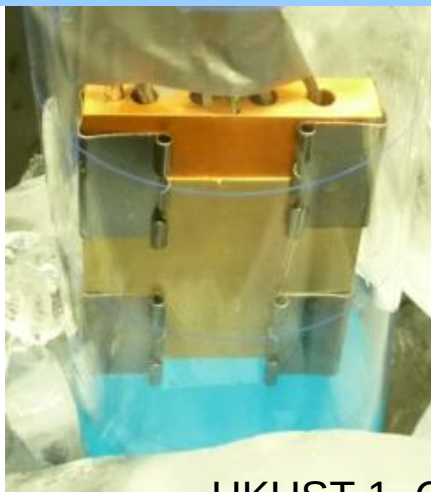
- structural defect interrupt the periodic arrangement
- defects in MOFs can be introduced
 - active sites on metal atoms
 - for microporous-mesoporous hierarchical structures
- different types of defects:



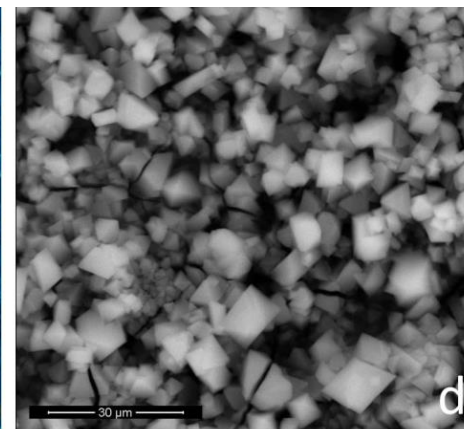
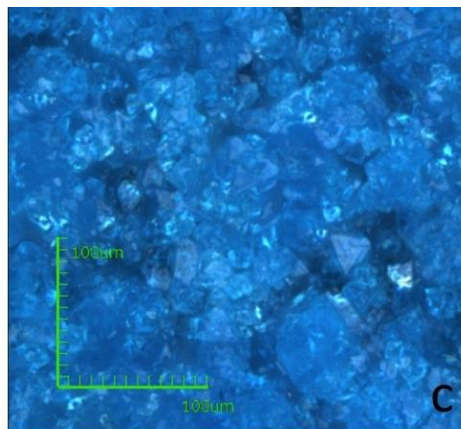
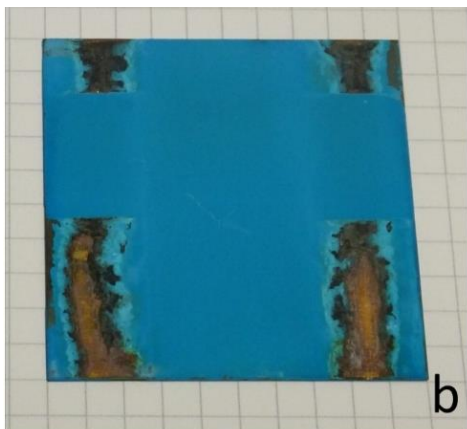
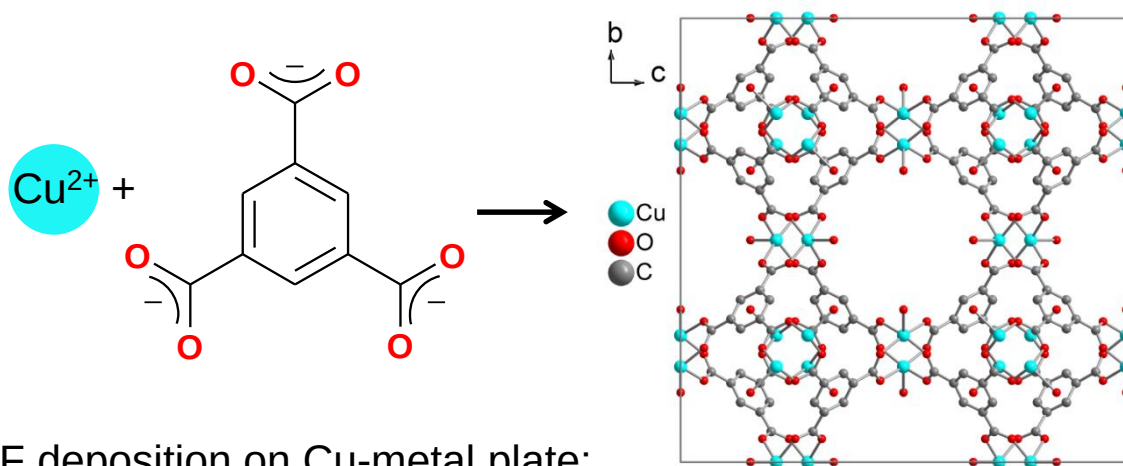
Z. Fang, B. Bueken, D. E. De Vos, R. A. Fischer, *Angew. Chem. Int. Ed.*, **2015**, 54, 7234–7254.
M. Vandichel, J. Hajek, F. Vermoortele, M. Waroquier, D. E. De Vos, V. Van Speybroeck, *CrystEngComm*, **2015**, 17, 395–406.
F. Vermoortele, B. Bueken, G. Le Bars, B. Van de Voorde, M. Vandichel, K. Houthoofd, A. Vimont, M. Daturi, M. Waroquier, V. Van Speybroeck, C. Kirschhock, D. E. De Vos, *J. Am. Chem. Soc.*, **2013**, 135, 11465–11468.

MOFs

- thin continuous film preparation as coating on metal sheet



HKUST-1, Cu-btc MOF deposition on Cu-metal plate:

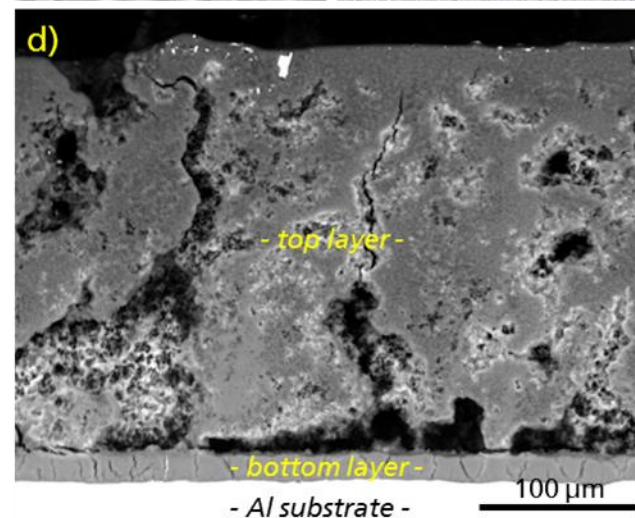
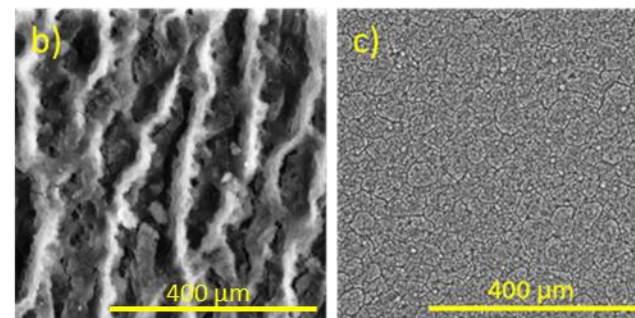
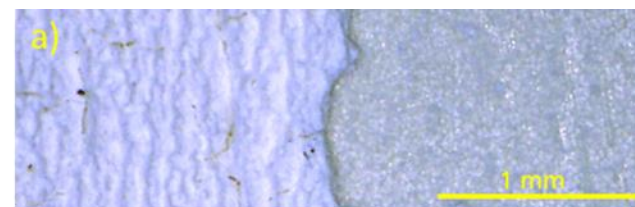
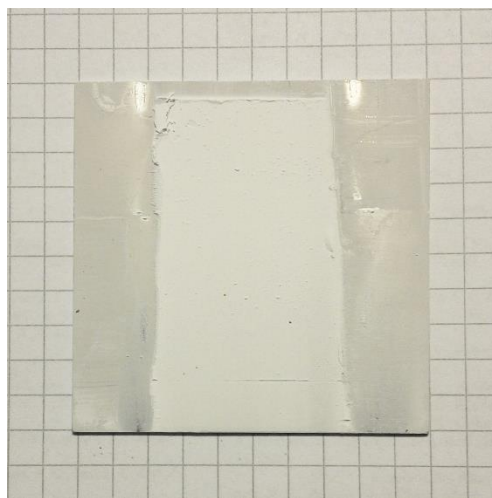
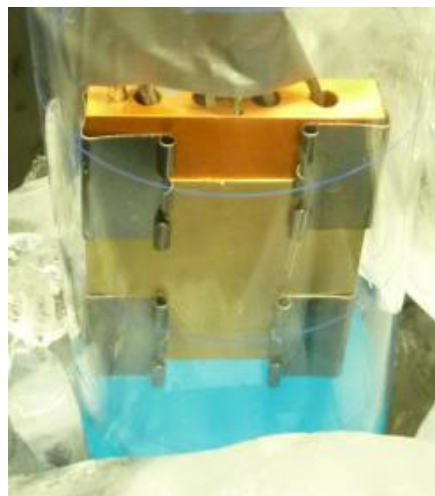
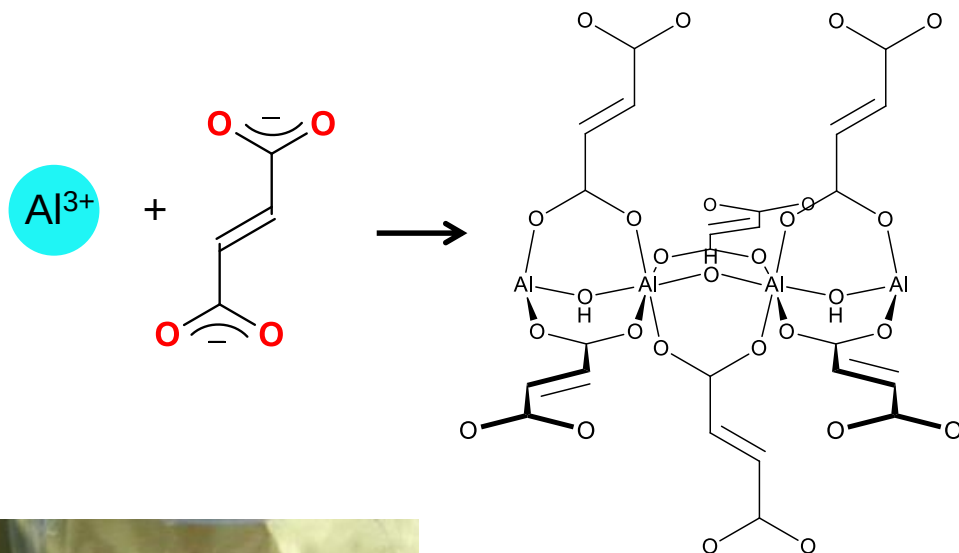


- polycrystalline, highly stable, thermally conductive MOF coating
- thermal conductivity 1.2 to $1.4 \text{ W m}^{-1} \text{ K}^{-1}$ > active carbon, silica gel
- $S(\text{BET})$: $1160 \text{ m}^2 / \text{g}$ (Lit.: $690\text{-}1500 \text{ m}^2 / \text{g}$)

MOFs

- thin continuous film preparation as coating on metal sheet

Aluminum fumarate MOF (Basolite™ A520) deposition on Al-metal plate:



MOF synthesis and analysis

- first time: crystal structure analysis

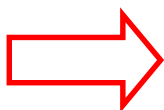
First time structure analysis by

– single crystal X-ray (synchrotron) structure analysis

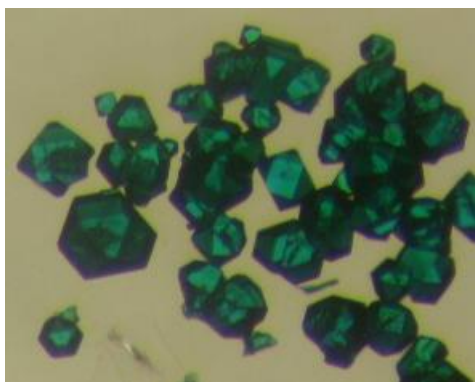
(also Rietveld refinement from X-ray powder diffraction)

⇒ structure analysis and discussion(!)
of metal-ligand architecture, supramolecular interactions etc.

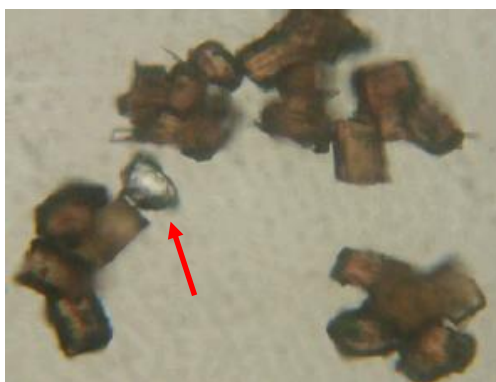
– before / together with visual microscopic inspection



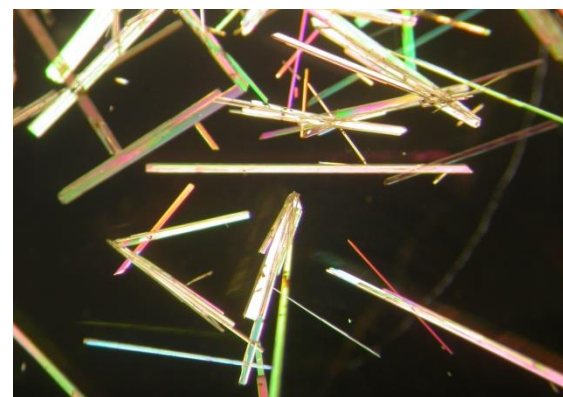
- homogeneity of (crystalline) product (shape, color)
- crystal quality (with polarizing microscope)



– homogeneous



– second phase
– manual separation possible?

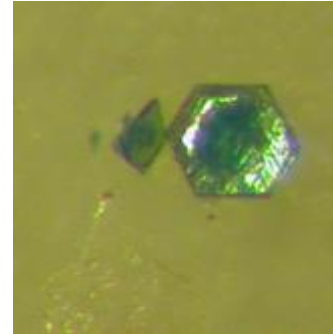
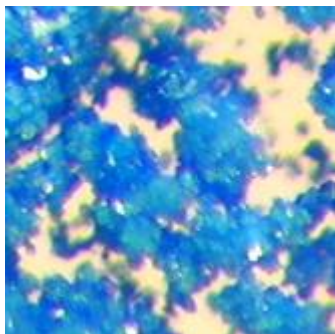
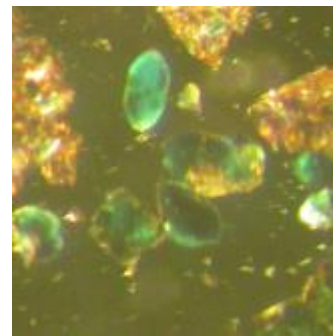
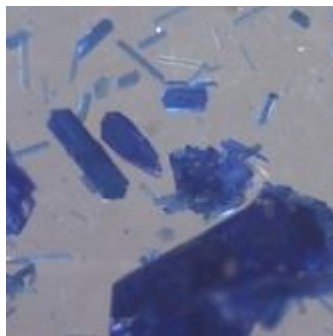


– homogeneous
– coloring indicates intergrowth

MOF synthesis and analysis

- first time: crystal structure analysis

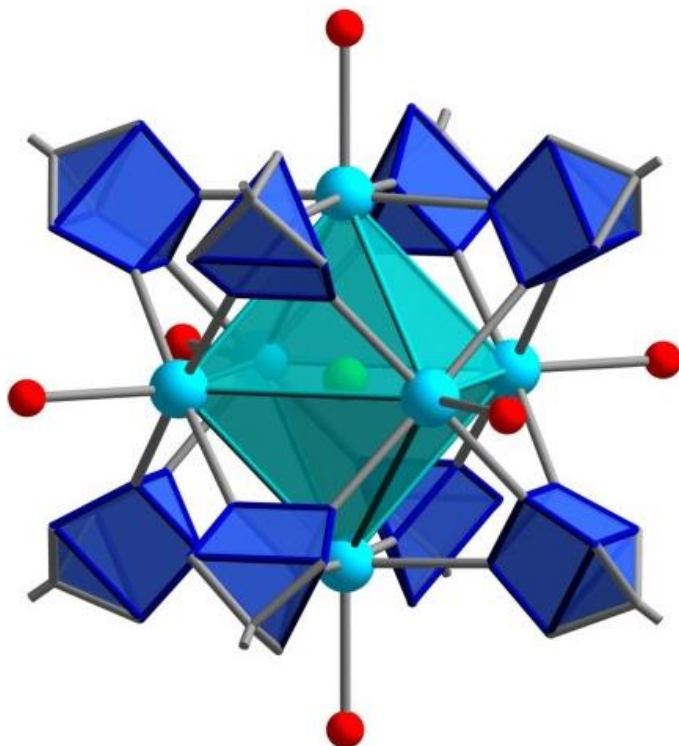
First time structure analysis by
– **single crystal X-ray (synchrotron) structure analysis**
(also Rietveld refinement from X-ray powder diffraction)



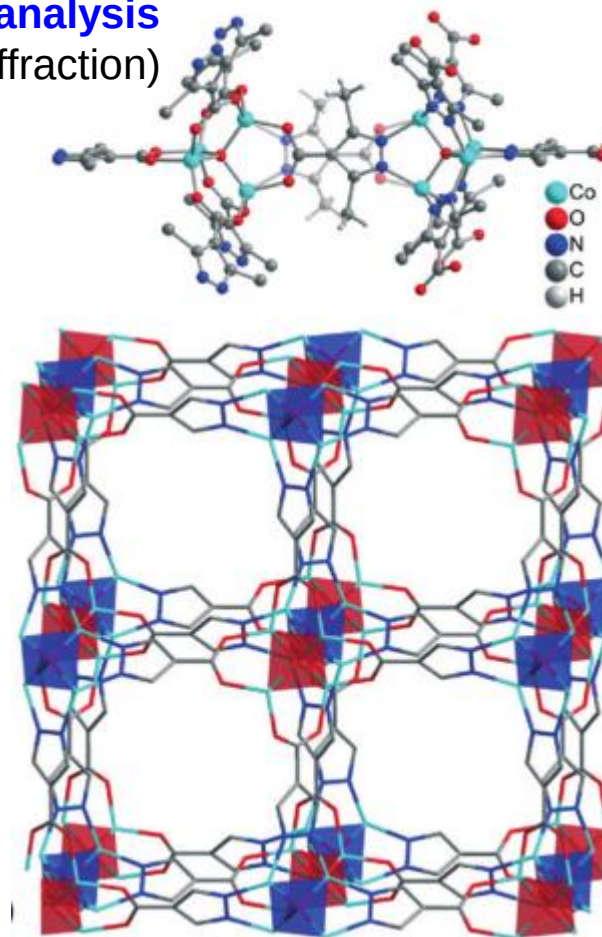
MOF synthesis and analysis

- first time: crystal structure analysis

First time structure analysis by
– single crystal X-ray (synchrotron) structure analysis
(also Rietveld refinement from X-ray powder diffraction)



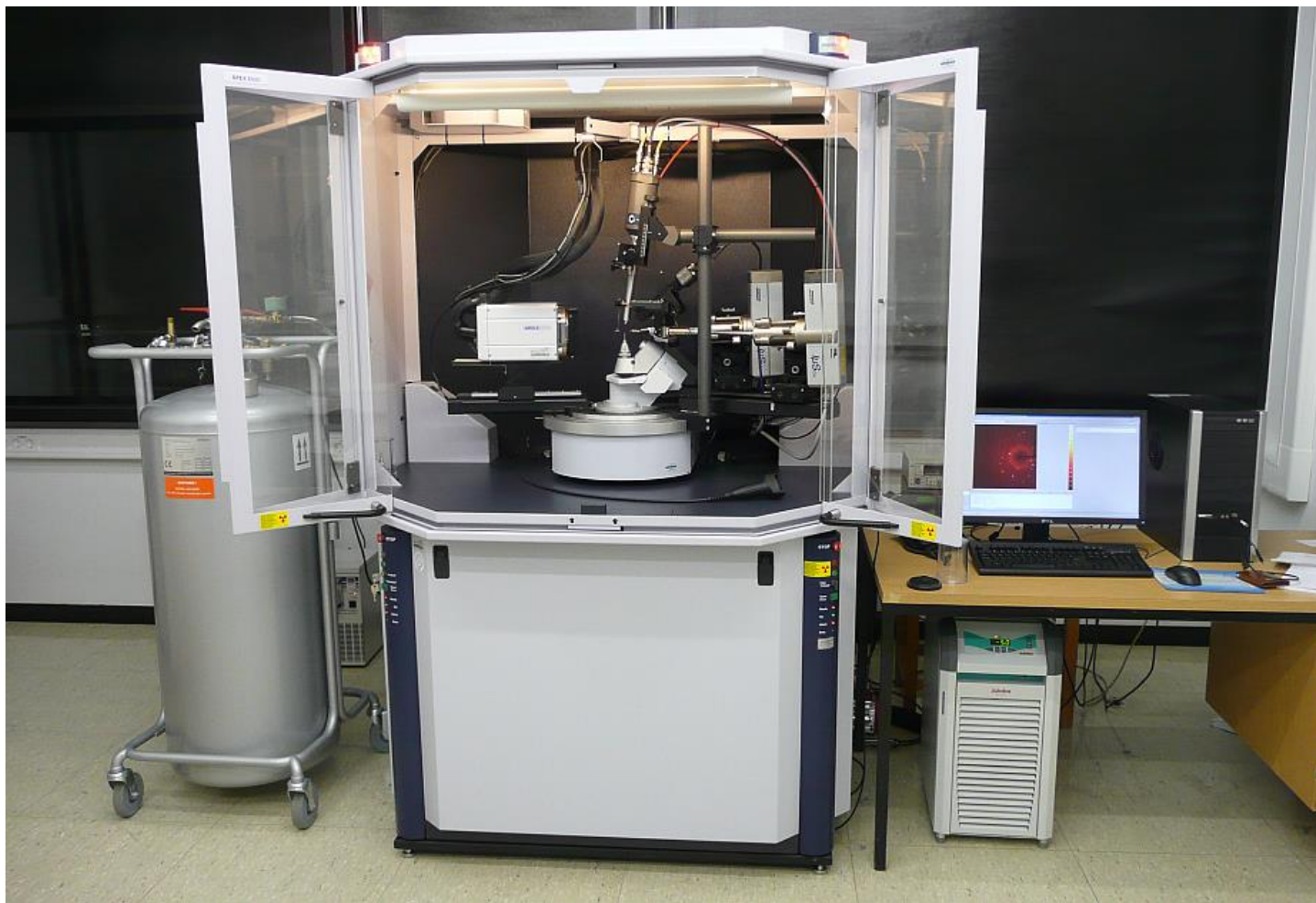
CrystEngComm **2014**, 16, 148-151.
DOI: 10.1039/C3CE42162G



CrystEngComm **2013**, 15, 9757. DOI: 10.1039/C3CE41426D

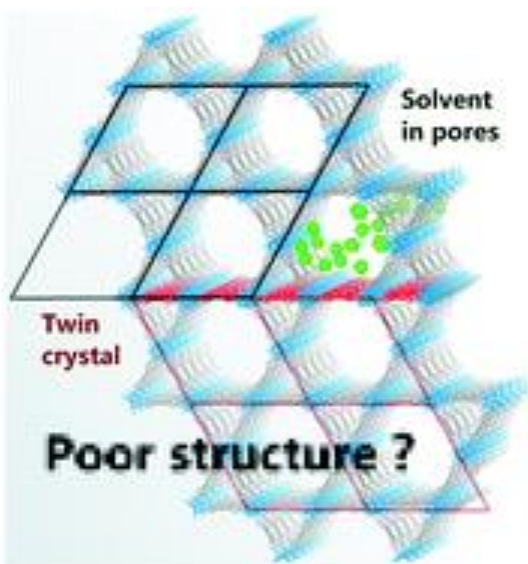
MOF structures - 3 case studies

Data collection: Bruker Kappa APEX DUO with APEX II CCD area detector and Mo-K α and Cu-K α microfocus sealed tubes





Cite this: *Chem. Soc. Rev.*, 2017, 46, 4867



Pitfalls in metal–organic framework crystallography: towards more accurate crystal structures†

S. Øien-Ødegaard, * G. C. Shearer, D. S. Wragg and K. P. Lillerud

Key learning points

- Single crystal X-ray diffraction is the most utilized and powerful tool for structure determination.
- The contents of a MOF's pores have large impact on its diffraction pattern due to scattering, and filled pores should thus be avoided during diffraction experiments.
- MOF crystals will in many cases degrade upon rapid solvent exchange or evacuation, and should thus be preserved during such postsynthetic treatments.
- Many MOFs are prone to twinning, and it may even be inevitable in certain cases during structure determination using the appropriate tools.
- Poor handling of samples and data can inhibit the determination of the true crystal structure, and in some cases incorrectly described structures.



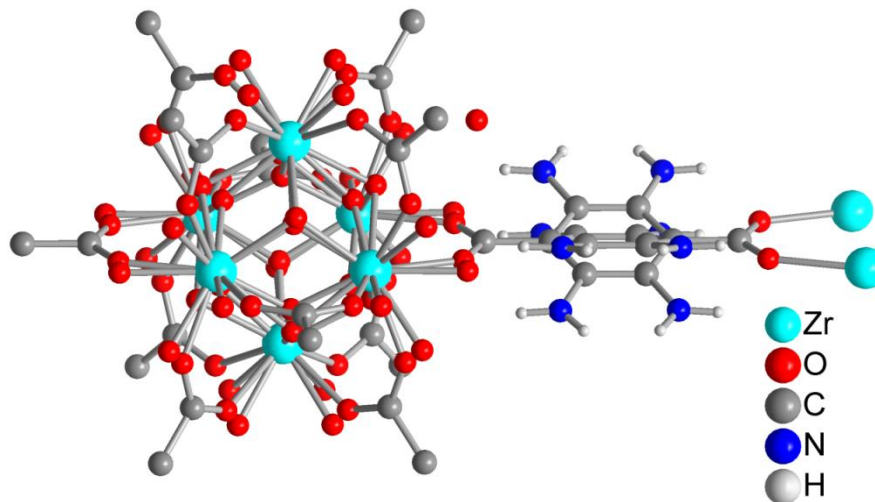
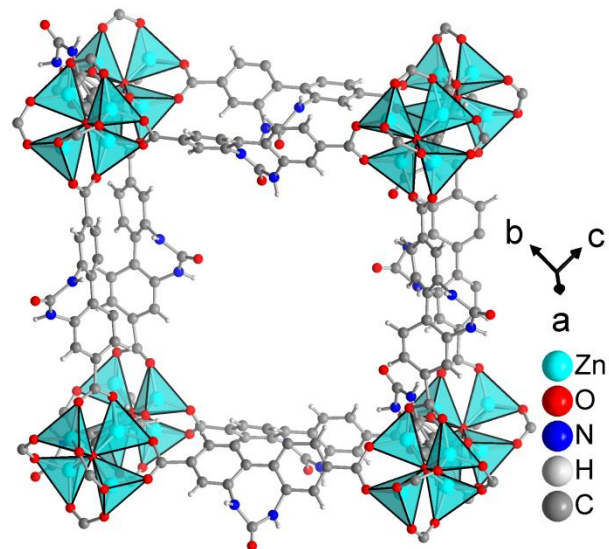
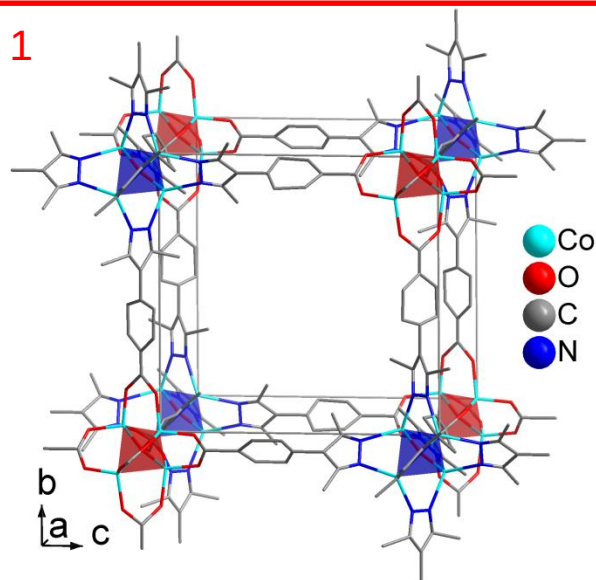
Cite this: *Chem. Soc. Rev.*, 2017,
46, 4867

Pitfalls in metal–organic framework crystallography: towards more accurate crystal structures†

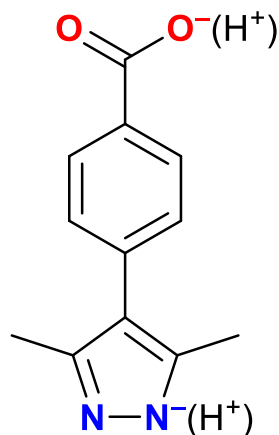
- contents of a MOF's pores have large impact on diffraction pattern due to scattering effects and guest/host interaction leading to local symmetry breaking, filled pores should thus be avoided during diffraction experiments.
- MOF crystals will in many cases degrade upon rapid solvent exchange or evacuation. To obtain high-quality crystallographic data, the crystallinity has to be preserved during such postsynthetic treatments.
- Many MOFs are prone to twinning, and it may even be inevitable in certain MOFs. However, twinning can be readily determined and taken into account during structure determination using the appropriate tools.
- Poor handling of samples and data can inhibit the determination of the true crystal symmetry, which in turn leads to crystal structures of poor quality and in some cases incorrectly described structures.

Our experience with MOF structures - 3 case studies

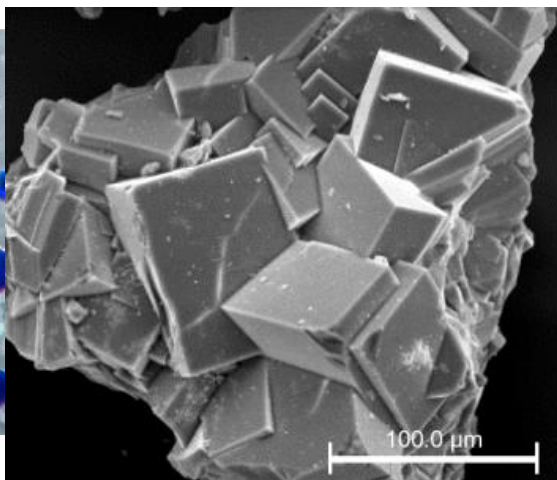
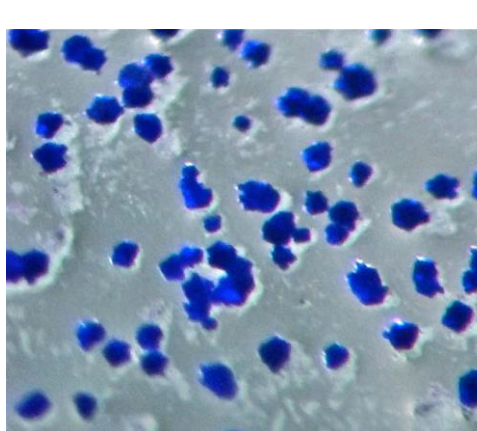
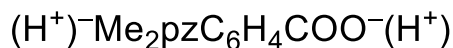
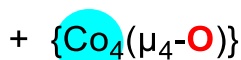
case 1



Mixed-donor coordination in $3D-[Co_4(\mu_4-O)(Me_2pzC_6H_4CO_2)_3]$ + ligand disorder



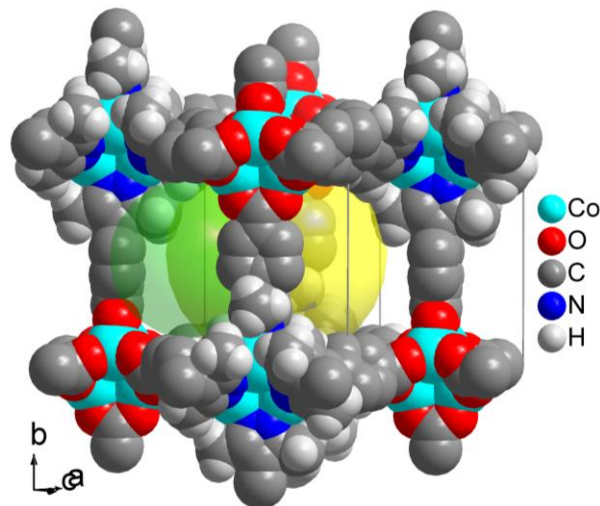
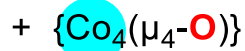
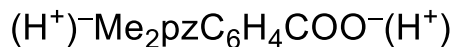
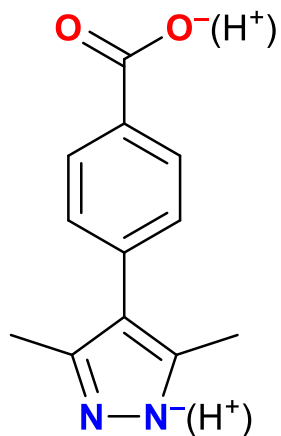
(+ small
crystal size)



Crystal system	orthorhombic
Space group	$P222$ (no. 16)
Temperature /K	100
Crystal size/mm	$0.01 \times 0.01 \times 0.01$
Radiation type	Cu-K α
θ range/ $^\circ$	$3.09 - 40.08$
$(\sin \theta/\lambda)_{\max} / \text{\AA}^{-1}$	0.418
$h; k; l$; range	$\pm 11; \pm 11; -10, 11$
$a/\text{\AA}$	14.242(3)
$b/\text{\AA}$	14.278(3)
$c/\text{\AA}$	14.2332(4)
$V/\text{\AA}^3$	2914.5(12)
Z	1
$D_{\text{calc}}/\text{g cm}^{-3}$	0.504
Reflect. collected (R_{int})	6548 (0.0890)
Independent reflections	1690
Parameters refined	130
R_1/wR_2 [$I > 2\sigma(I)$] ^e	0.0691 / 0.1453
R_1/wR_2 (all reflect.) ^e	0.1120 / 0.1574
Goodness-of-fit on F^2 ^f	0.923
Void electron count/e	749
Void, solvent accessible volume/ $\text{\AA}^3$ (% unit cell volume)	2107 (72%)

C. Heering, I. Boldog, V. Vasylyeva, J. Sanchiz, C. Janiak,
Bifunctional pyrazolate-carboxylate ligands for isorecticular cobalt and zinc MOF-5 analogs with magnetic analysis of the $\{Co_4(\mu_4-O)\}$ node;
CrystEngComm **2013**, 15, 9757-9768.

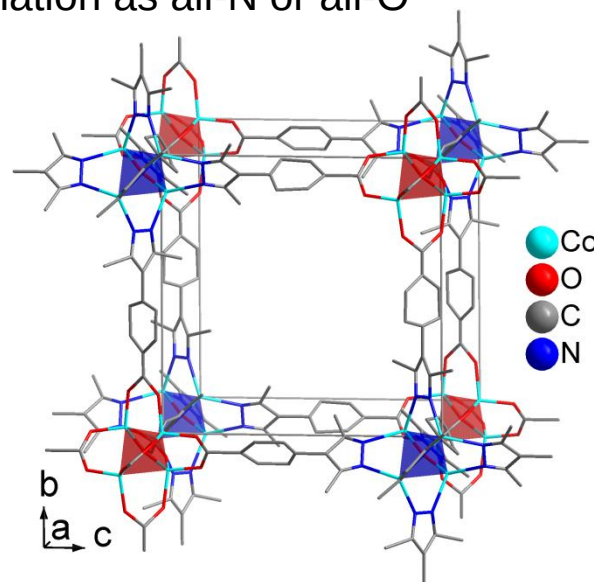
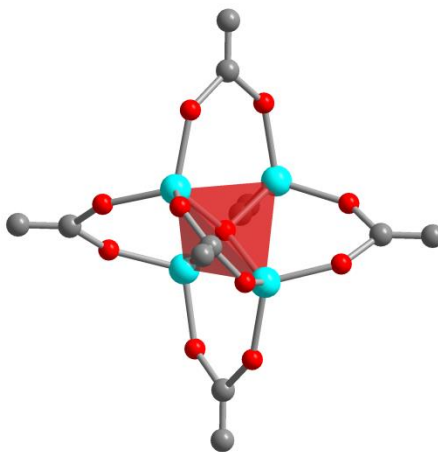
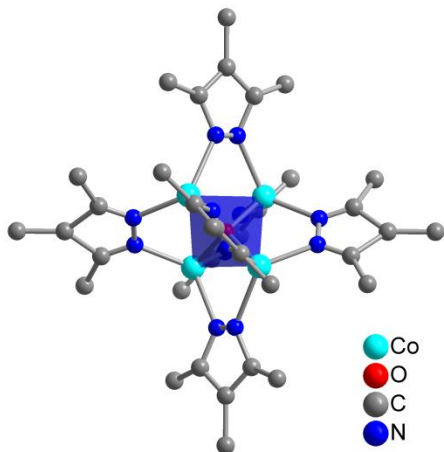
Mixed-donor coordination in $3D-[Co_4(\mu_4-O)(Me_2pzC_6H_4CO_2)_3]$ + ligand disorder



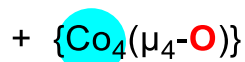
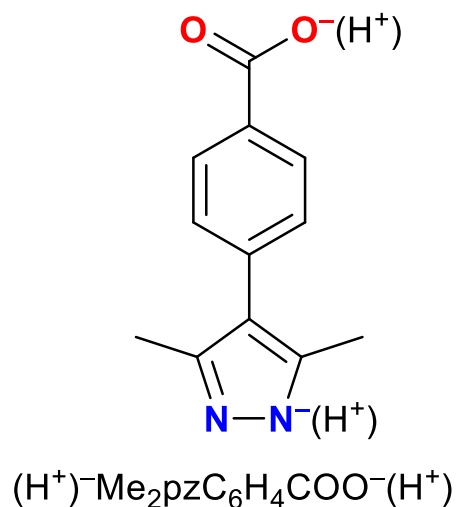
pcu-a network

yellow sphere, $\varnothing = 14 \text{ \AA}$
green sphere, $\varnothing = 10 \text{ \AA}$

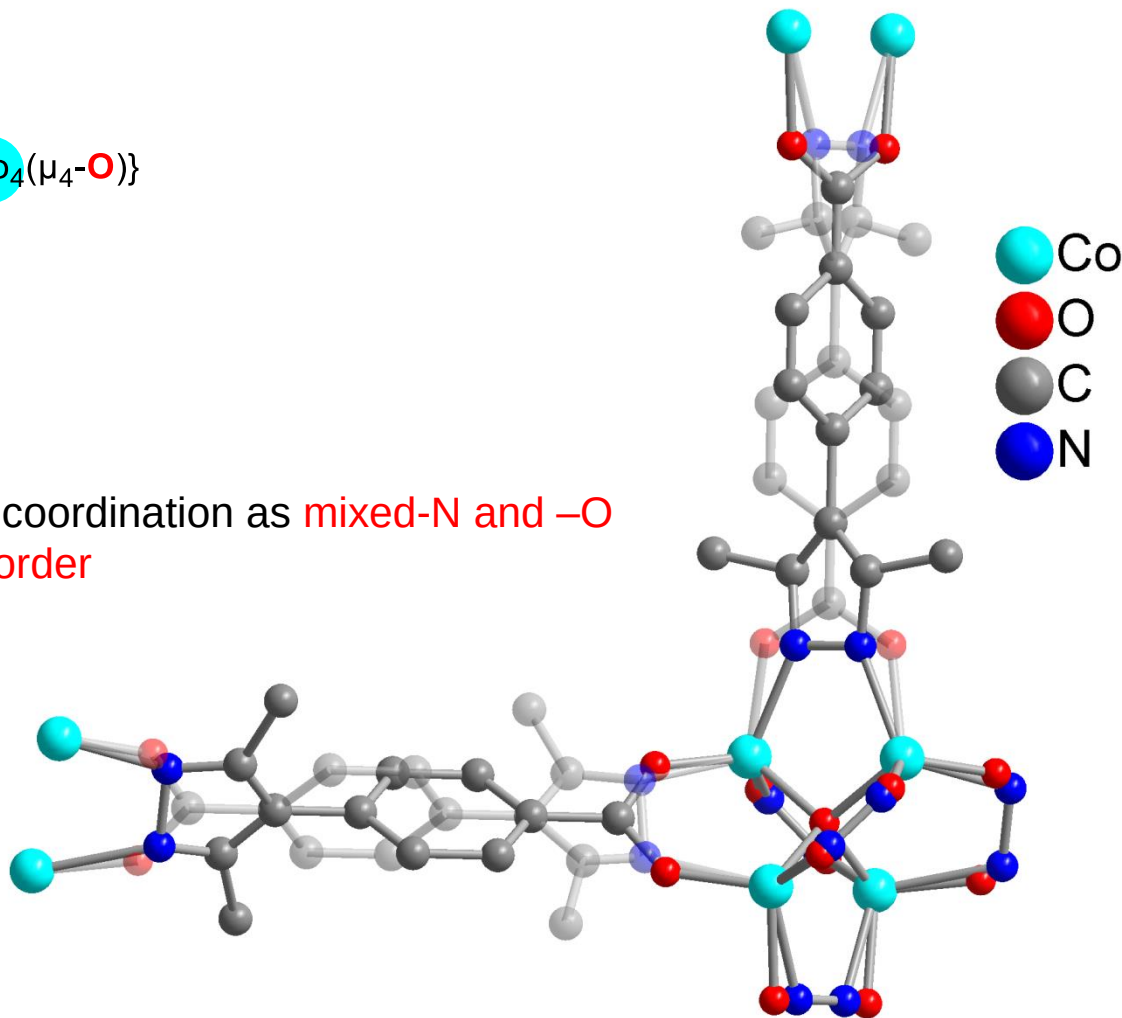
idealized structure: showing $Co_4(\mu_4-O)$ coordination as all-N or all-O



Mixed-donor coordination in $3D-[Co_4(\mu_4-O)(Me_2pzC_6H_4CO_2)_3]$ + ligand disorder

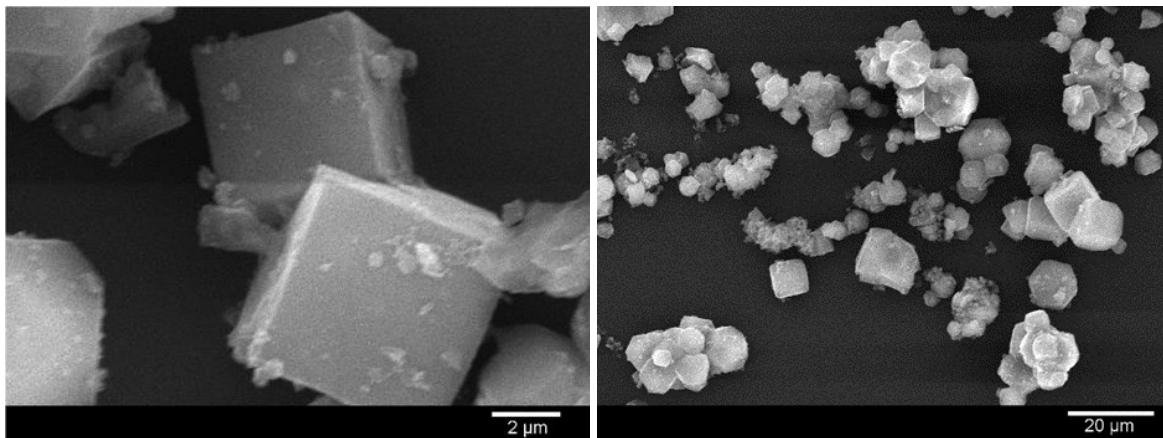


real structure: $Co_4(\mu_4-O)$ coordination as mixed-N and $-O$
- together with ligand disorder



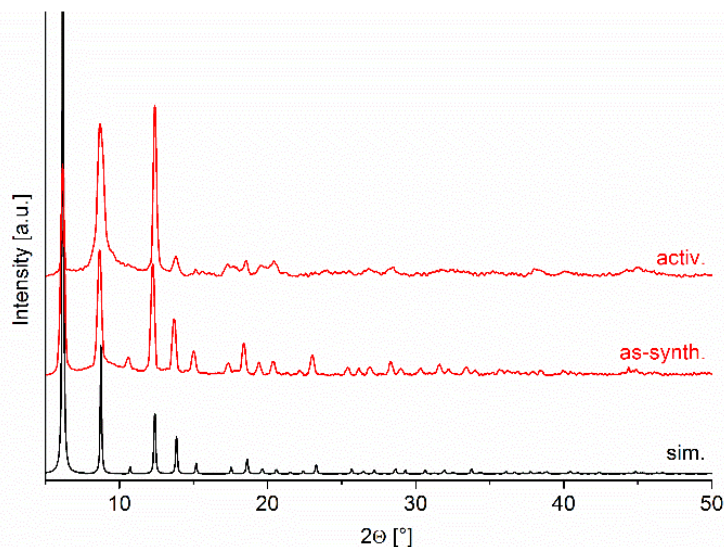
Mixed-donor coordination in $3D-[Co_4(\mu_4-O)(Me_2pzC_6H_4CO_2)_3]$ + ligand disorder

new microwave synthesis yielded less aggregated crystals:

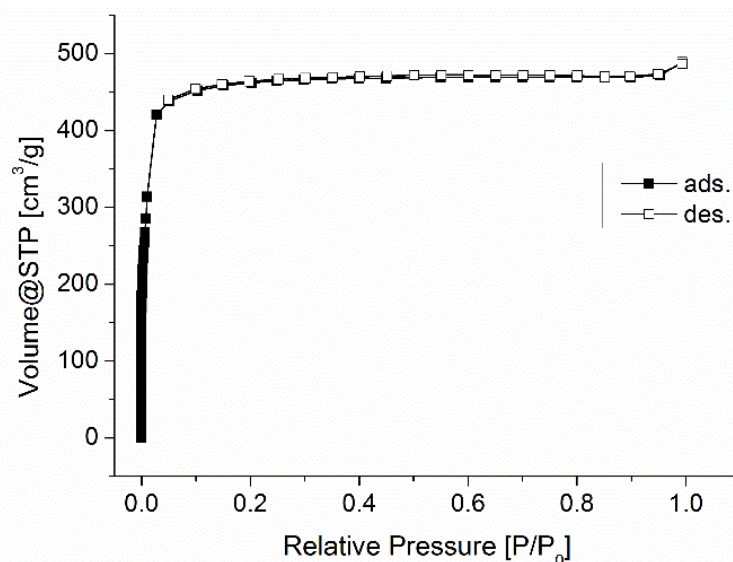


J. Dechnik, A. Nuhnen, C. Janiak,
Mixed-matrix membranes of the
air-stable MOF-5 analog $[Co_4(\mu_4-O)(Me_2pzba)_3]$ with mixed-
functional pyrazolate-carboxylate
linker for CO_2/CH_4 separation;
Cryst. Growth Des. **2017**, 17,
4090-4099.

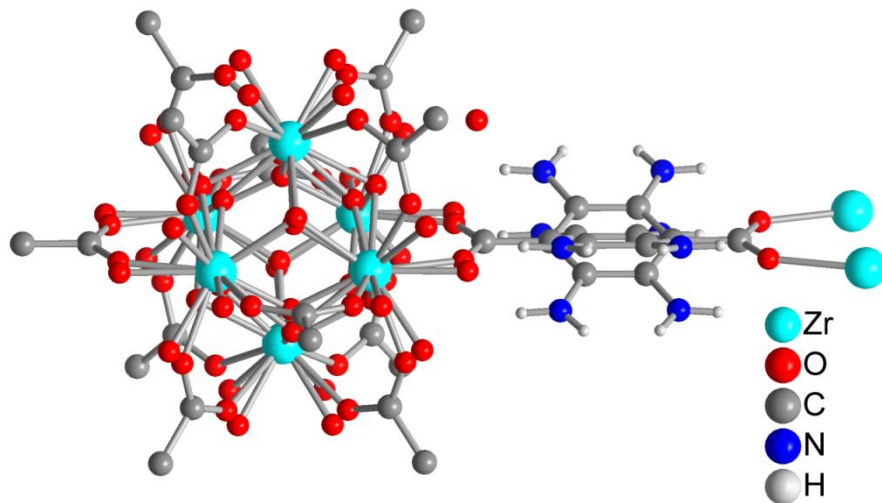
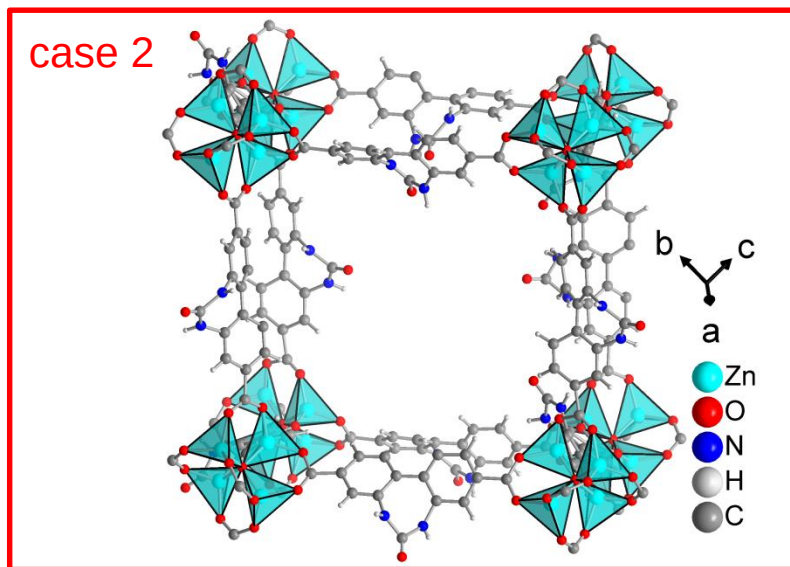
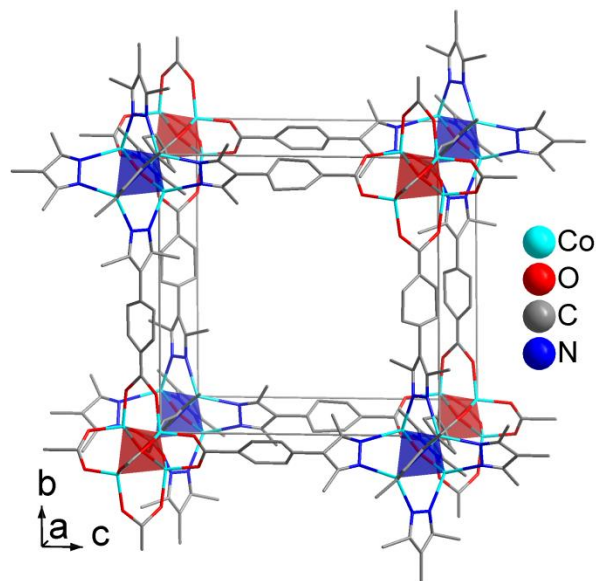
PXRD:



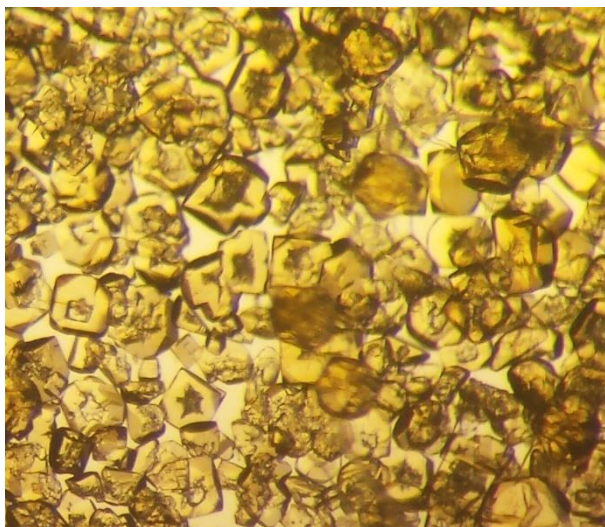
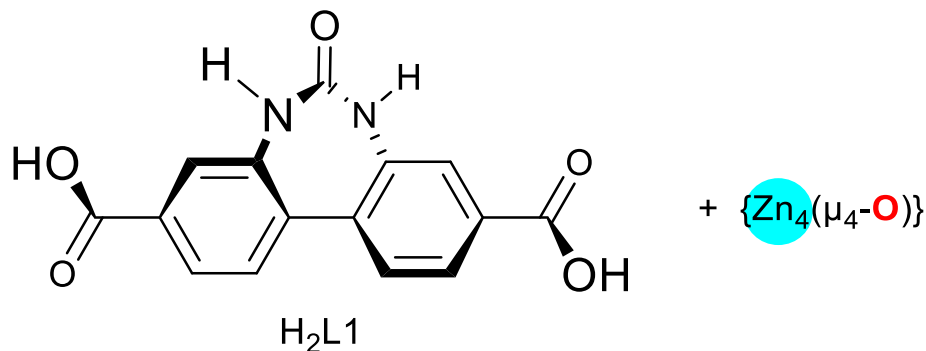
N_2 -BET: 1850 m²/g



Our experience with MOF structures - 3 case studies



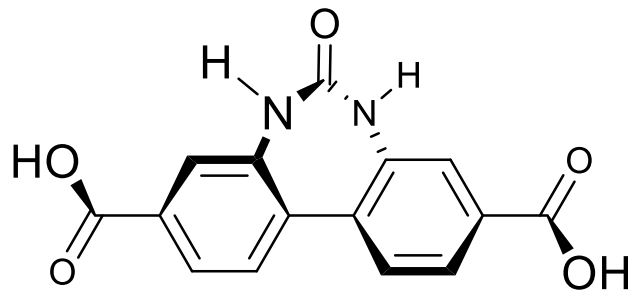
Ligand disorder in $3D\text{-}[\text{Zn}_4(\mu_4\text{-O})(\text{L1})_3]$ + interpenetration



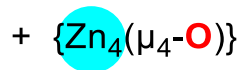
Crystal system, SG	Trigonal, $R\bar{3}c:H$
Temperature (K)	100
Crystal size (mm)	$0.17 \times 0.10 \times 0.08$
Radiation type	Cu-K α
θ range/ $^\circ$	2.23 – 54.57
$(\sin \theta/\lambda)_{\text{max}}$ ($\text{\AA}^{-1}$)	0.528
$a=b/(\text{\AA})$	24.4635 (15)
$c/(\text{\AA})$	112.996 (8)
$V/(\text{\AA}^3)$	58564 (8)
Z	12
$D_{\text{calc}}/\text{g cm}^{-3}$	0.792
No. of measured, indep., obs. [$I > 2\sigma(I)$] reflections	65987, 8018, 4521
R_{int}	0.081
Data/parameters	8018/464
R_1/wR_2 [$I > 2\sigma(I)$]	0.1107 / 0.3052,
R_1/wR_2 (all data)	0.1564 / 0.3443
Goodness-of-fit on F^2	1.100
Void electron count/e	10520
Void, solvent accessible volume/ $\text{\AA}^3$ (% unit cell volume)	37071 (63%)

S. Glomb, D. Woschko, G. Makhloufi, C. Janiak,
Metal-Organic Frameworks with Internal Urea-Functionalized Dicarboxylate Linkers for SO_2 and NH_3 Adsorption;
ACS Appl. Mater. Interfaces **2017**, 9, 37419-37434.

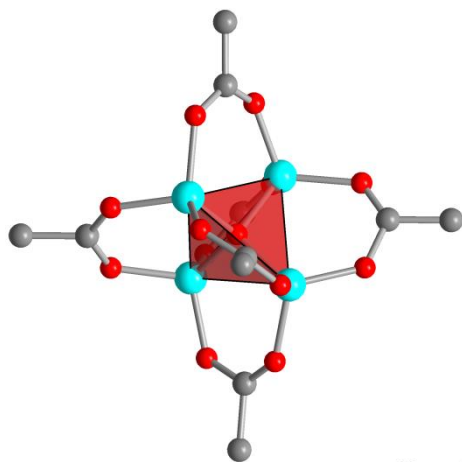
Ligand disorder in 3D-[Zn₄(μ₄-O)(L1)₃] + interpenetration



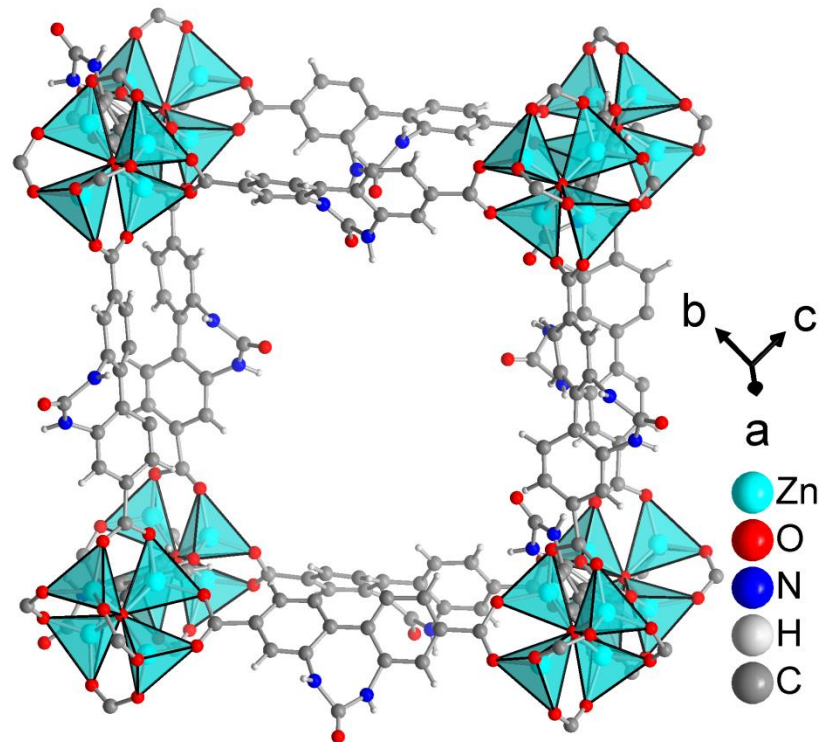
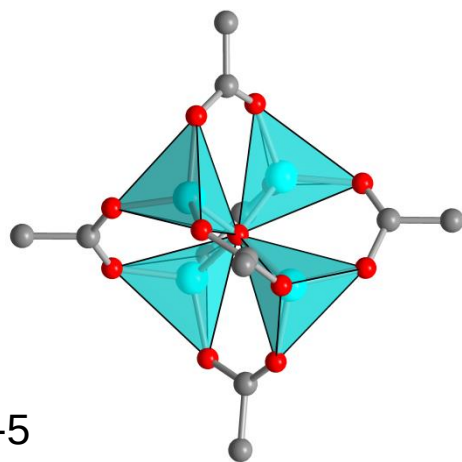
H₂L1



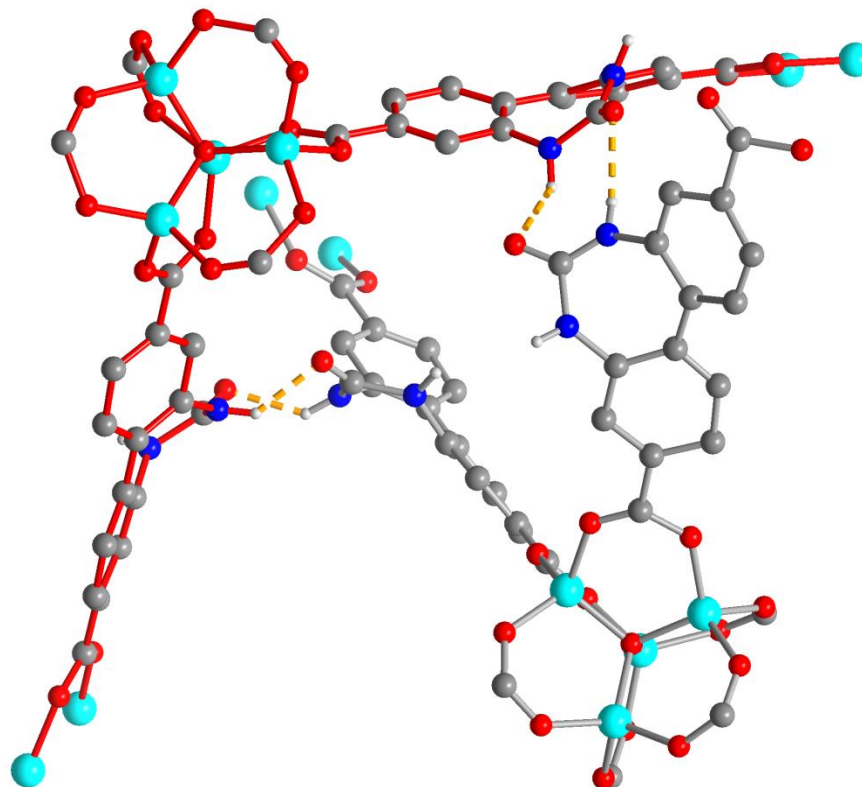
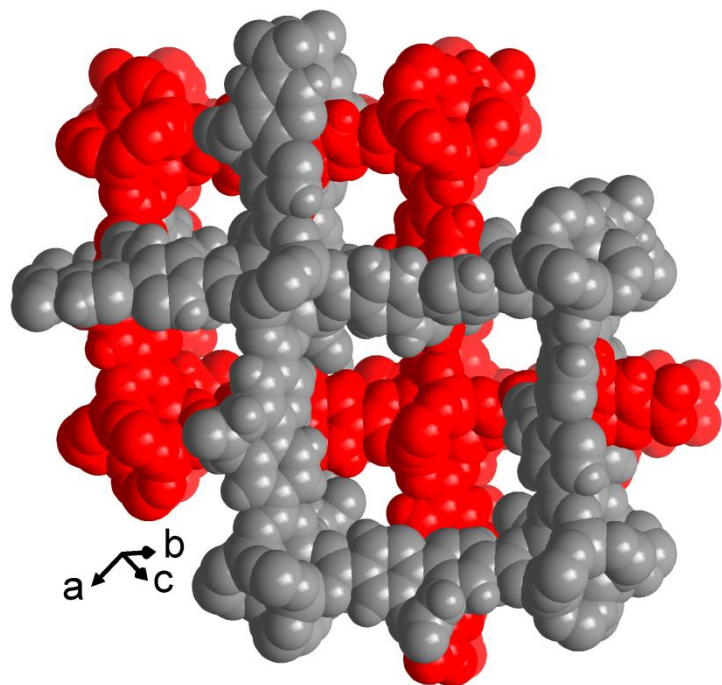
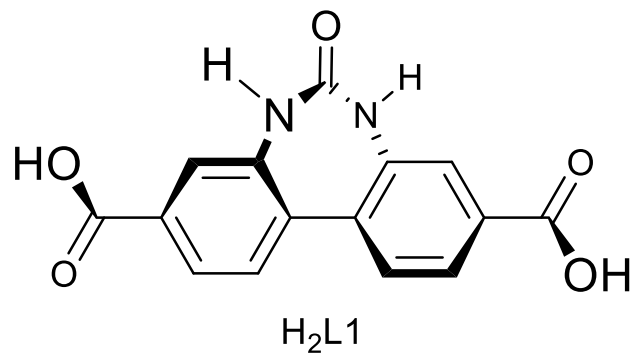
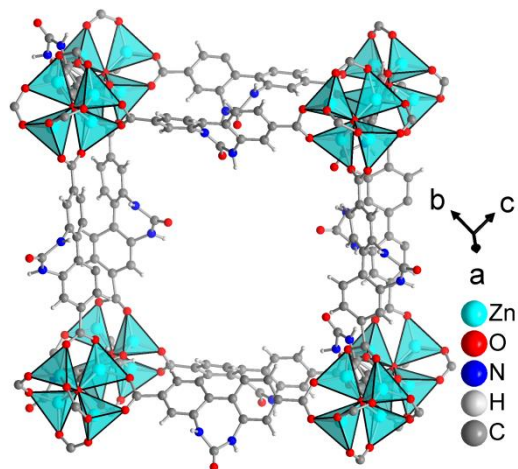
pcu-a network



SBU as in MOF-5

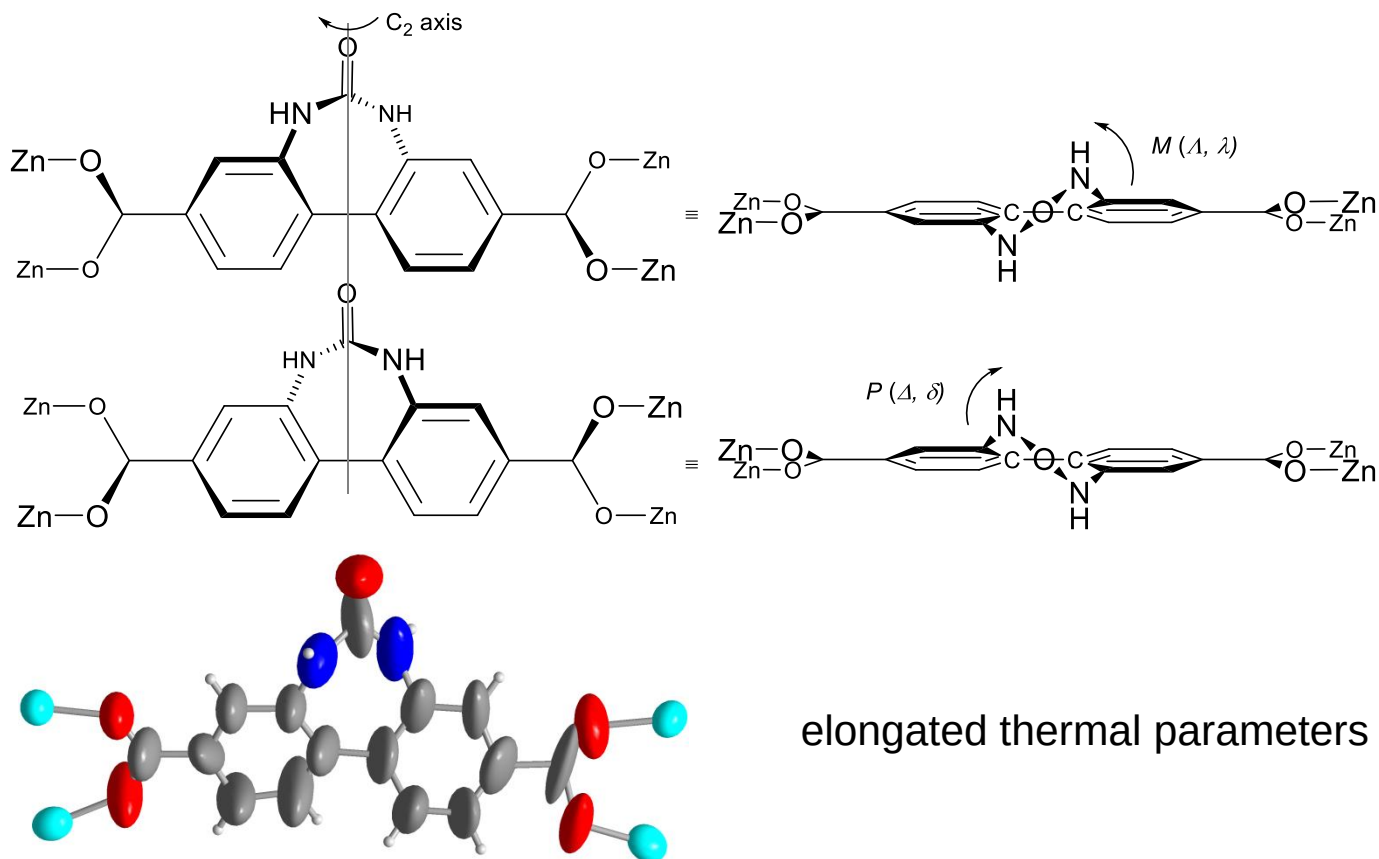


Ligand disorder in 3D-[Zn₄(μ₄-O)(L1)₃] + interpenetration



Ligand disorder in 3D-[Zn₄(μ₄-O)(L1)₃] + interpenetration

non-statistical twist-, oscillating or racemizing disorder over both chiral configurations:

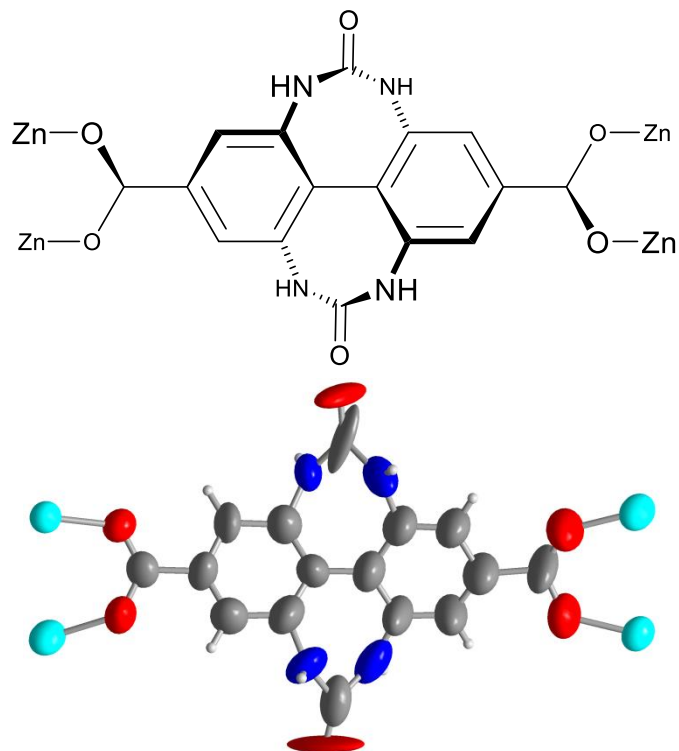


elongated thermal parameters

S. Glomb, D. Woschko, G. Makhloufi, C. Janiak,
Metal-Organic Frameworks with Internal Urea-Functionalized Dicarboxylate Linkers for SO₂ and NH₃ Adsorption;
ACS Appl. Mater. Interfaces **2017**, 9, 37419-37434.

Ligand disorder in $3D-[Zn_4(\mu_4-O)(L1)_3]$ + interpenetration

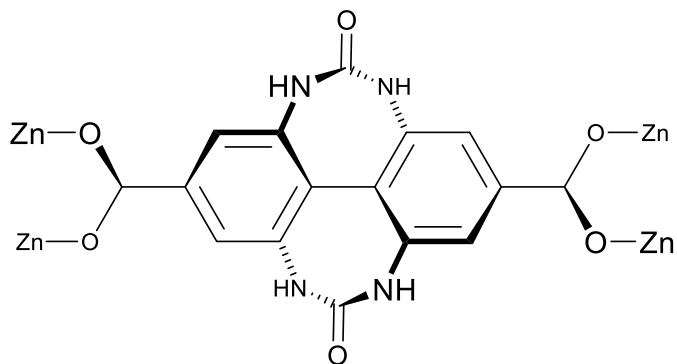
orientation disorder of the urea group:



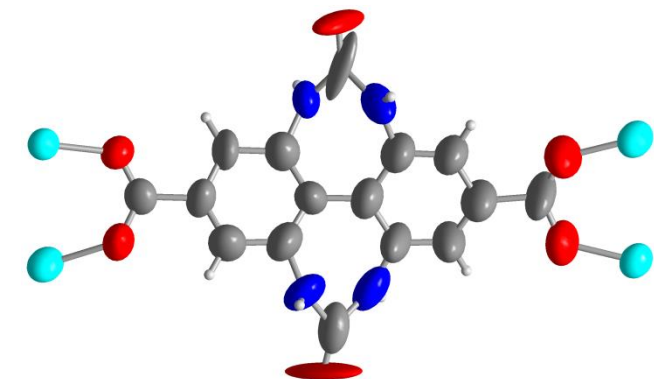
NH-CO-NH atoms
only half-occupied

Ligand disorder in $3D-[Zn_4(\mu_4-O)(L1)_3]$ + interpenetration

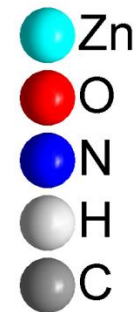
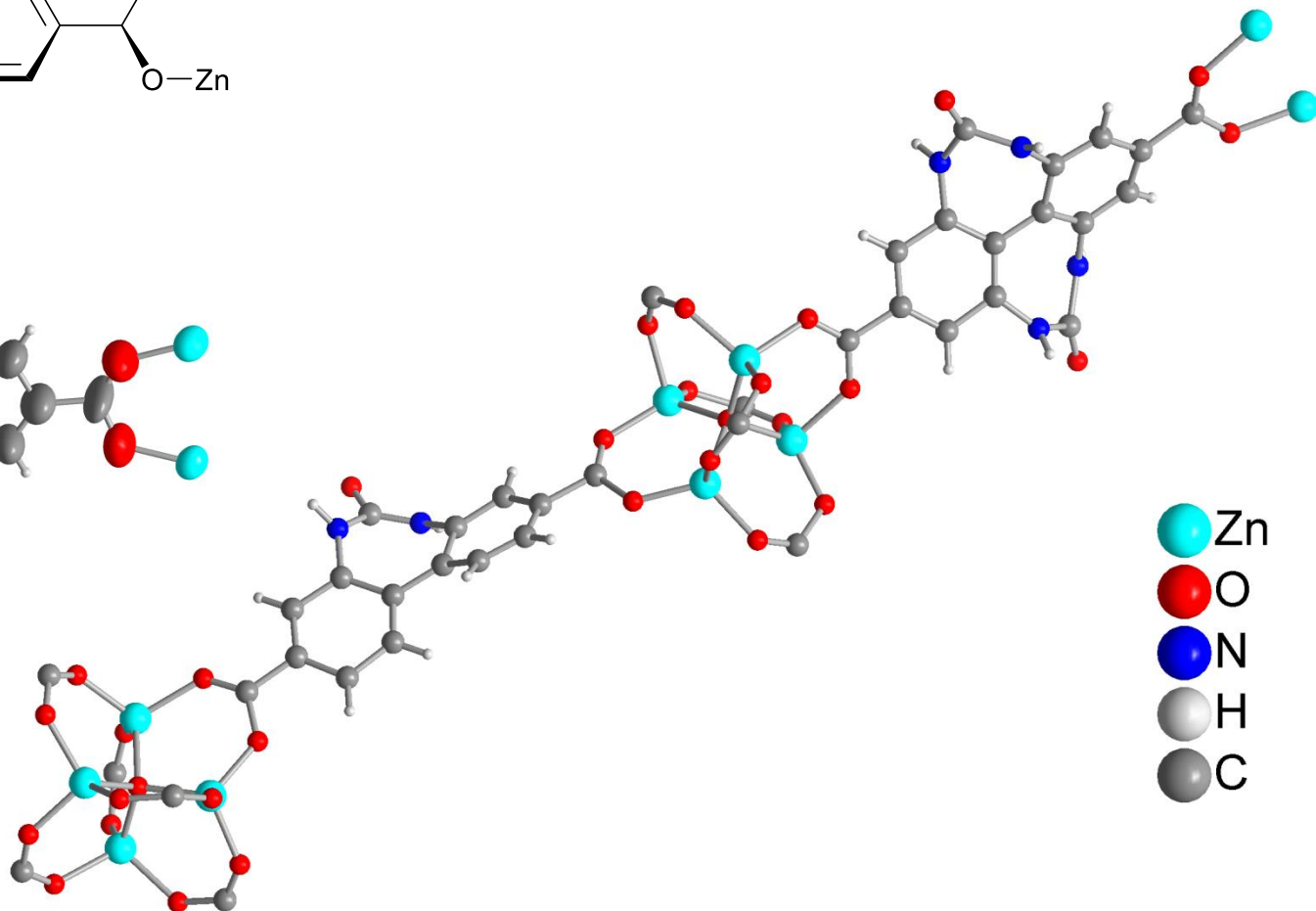
orientation disorder of the urea group:



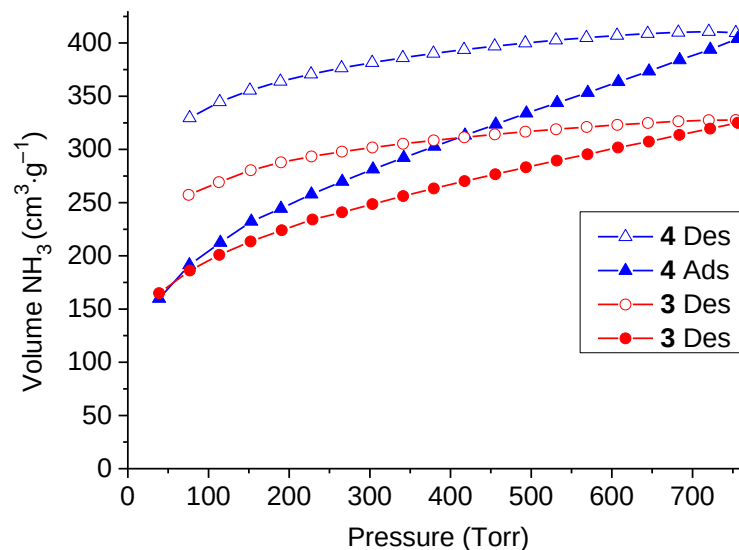
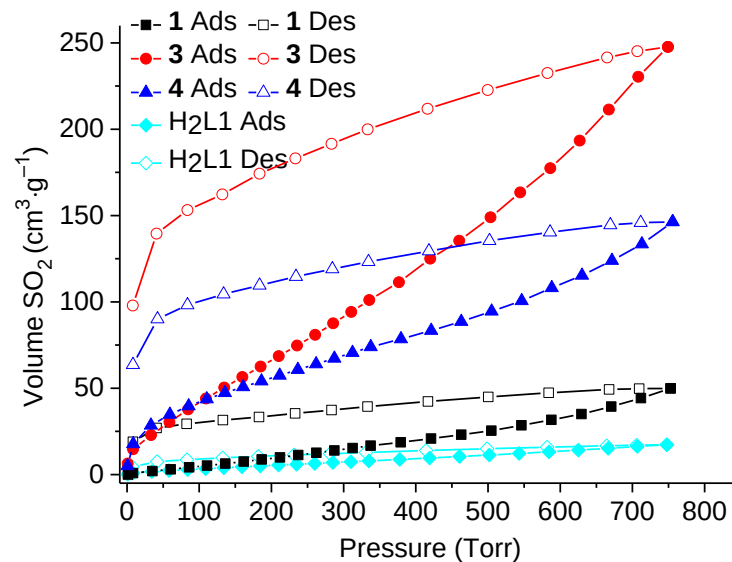
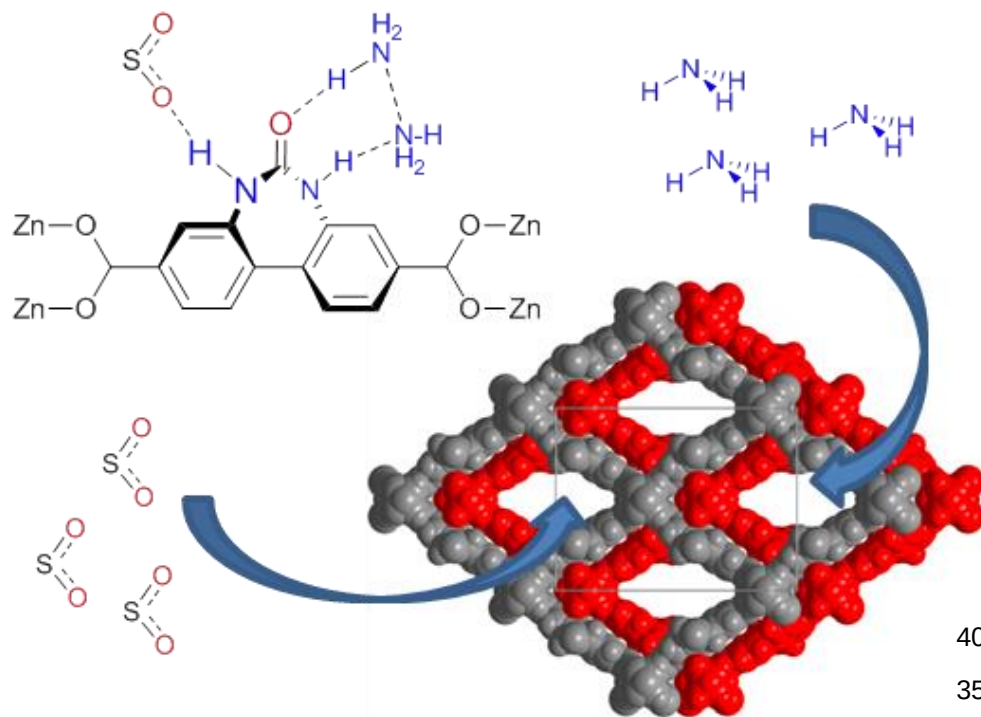
⇒ two crystallographically different linkers $L1^{2-}$



NH-CO-NH atoms
only half-occupied

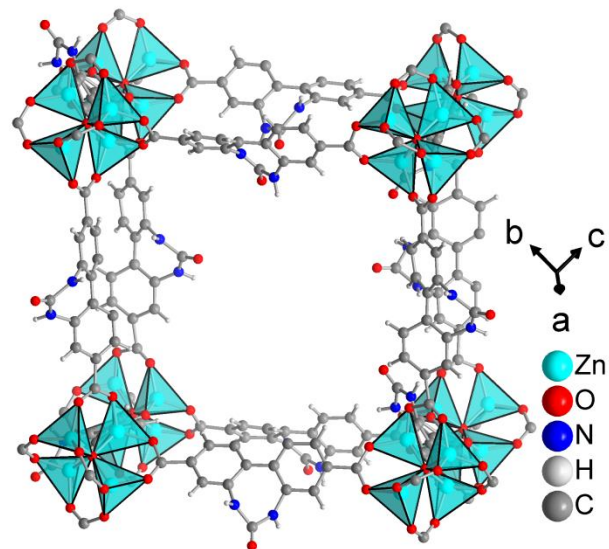
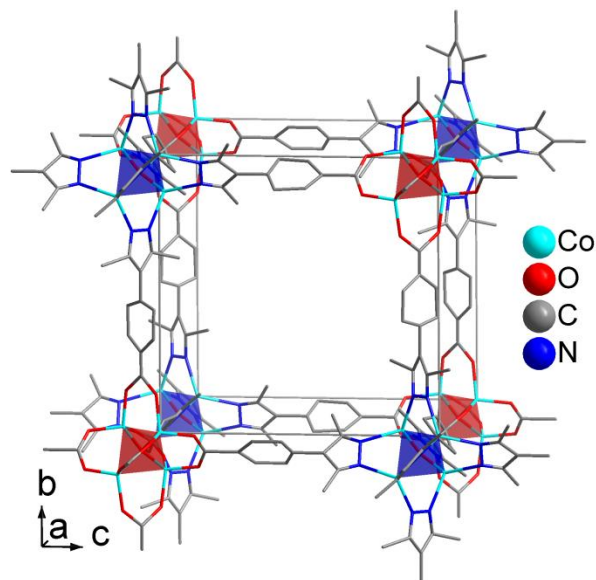


The idea behind using urea-functionalized ligands in 3D-[Zn₄(μ₄-O)(L1)₃] and related MOFs

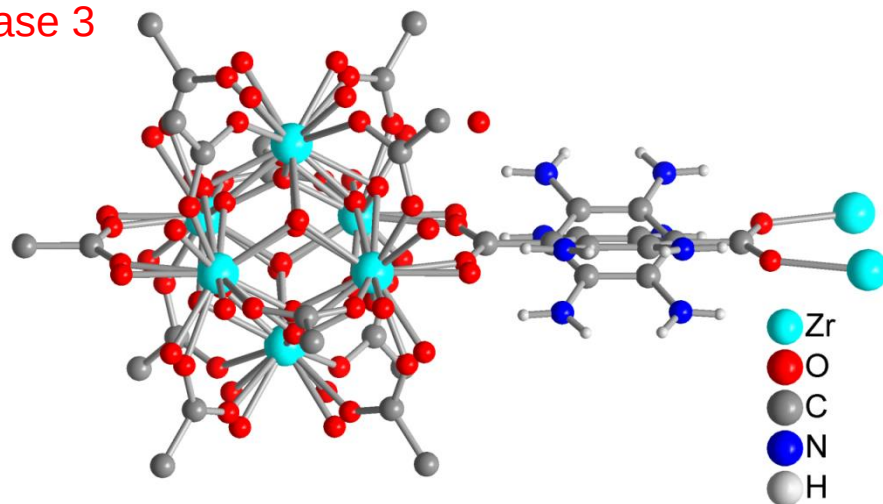


S. Glomb, D. Woschko, G. Makhloufi, C. Janiak,
Metal-Organic Frameworks with Internal Urea-
Functionalized Dicarboxylate Linkers for SO₂ and NH₃
Adsorption;
ACS Appl. Mater. Interfaces **2017**, 9, 37419-37434.

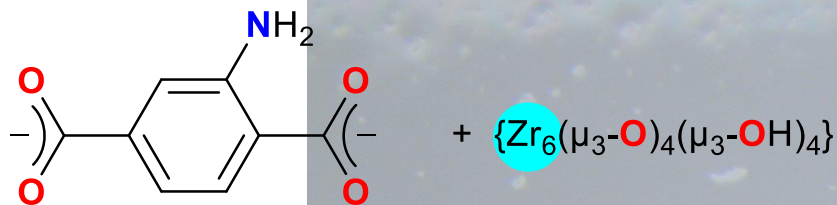
Our experience with MOF structures - 3 case studies



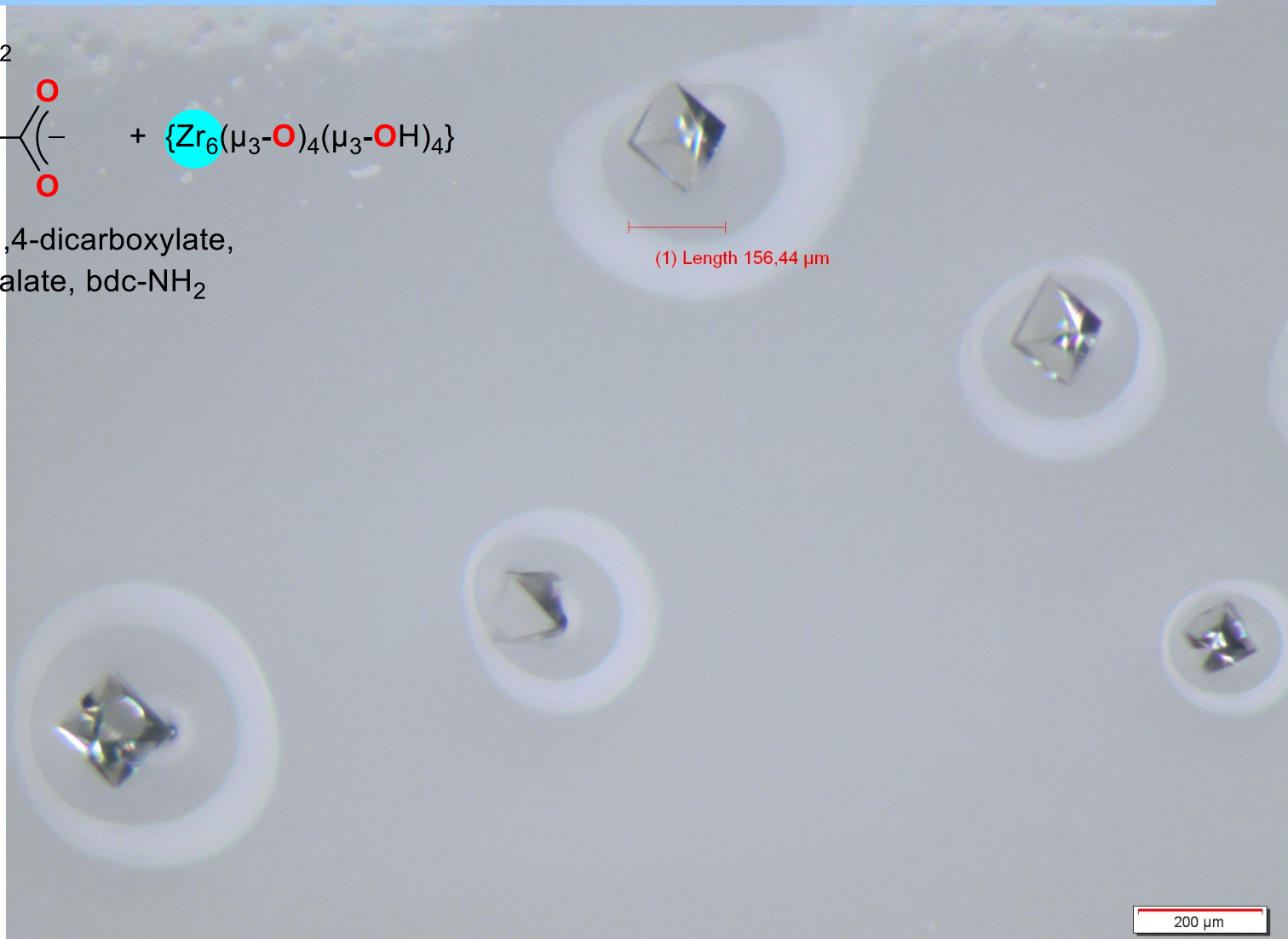
case 3



Linker disorder and defects in 3D-[Zr₆O₄(OH)₄(bdc+NH₂)₆] or UiO-66-NH₂

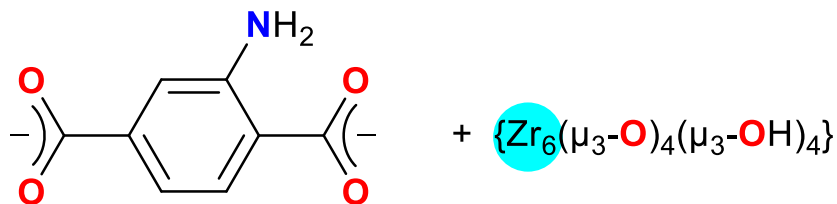


2-aminobenzene-1,4-dicarboxylate,
2-aminoterephthalate, bdc-NH₂



original publication – synchrotron structure: C. A. Trickett, K. J. Gagnon, S. Lee, F. Gándara, H.-B. Bürgi, O. M. Yaghi,
Definitive Molecular Level Characterization of Defects in UiO-66 Crystals.
Angew. Chem. Int. Ed. **2015**, 54, 11162 –11167.

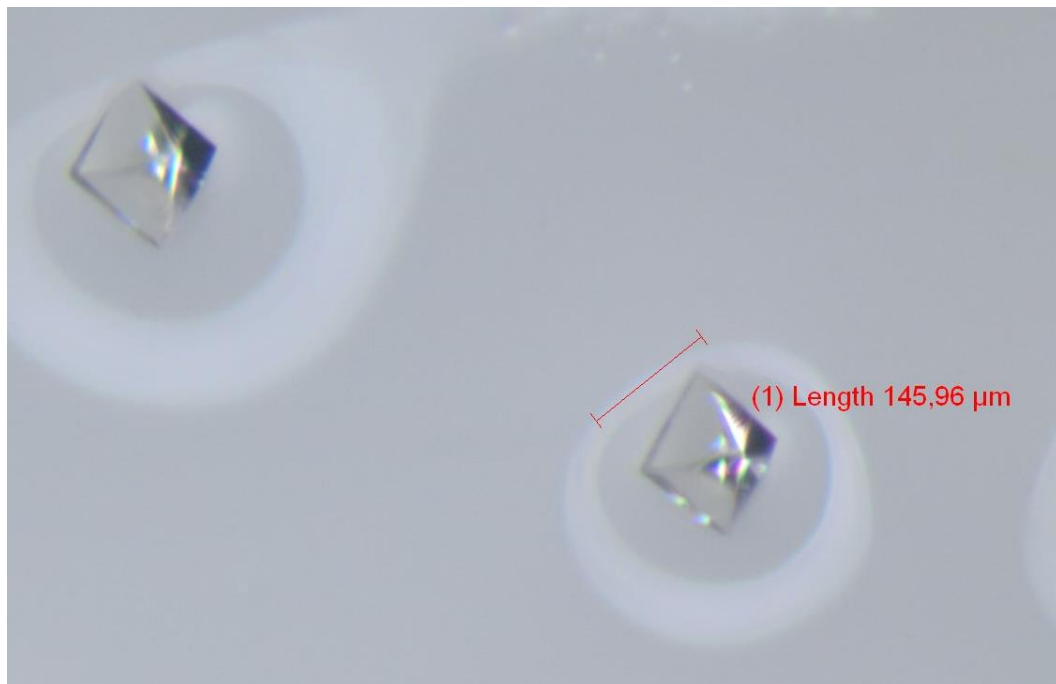
Linker disorder and defects in 3D-[Zr₆O₄(OH)₄(bdc+NH₂)₆] or UiO-66-NH₂



2-aminobenzene-1,4-dicarboxylate,
2-aminoterephthalate, bdc-NH₂

(redetermination – X-ray)

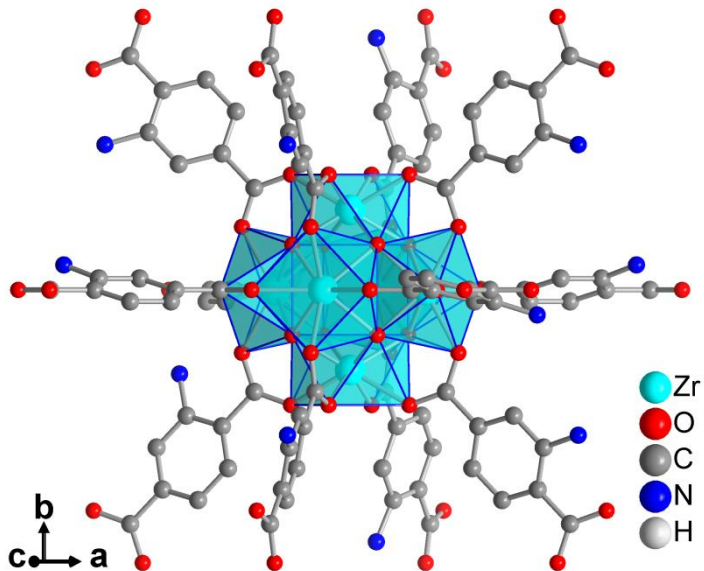
Crystal system	cubic
Space group	F m $\bar{3}$ m (no. 225)
Temperature /K	100
Crystal size/mm	0.1 × 0.1 × 0.1
Radiation type	Mo-K α
2 θ range/°	3.40 - 55.03
<i>h</i> ; <i>k</i> ; <i>l</i> ; range	–27,26; –23,27; –27,26
<i>a</i> /Å	20.799(1)
<i>V</i> /Å ³	8997(1)
<i>Z</i>	4
<i>D</i> _{calc} /g cm ^{–3}	1.234
Reflect. collected (<i>R</i> _{int})	47708 (0.048)
Independent reflections	581
Parameters refined	47
<i>R</i> ₁ / <i>wR</i> ₂ [<i>I</i> > 2 σ (<i>I</i>)] ^e	0.0386 / 0.1180
<i>R</i> ₁ / <i>wR</i> ₂ (all reflect.) ^e	0.0413 / 0.1230
Goodness-of-fit on <i>F</i> ² ^f	1.134
no squeeze needed	



original publication – synchrotron structure: C. A. Trickett, K. J. Gagnon, S. Lee, F. Gándara, H.-B. Bürgi, O. M. Yaghi, Definitive Molecular Level Characterization of Defects in UiO-66 Crystals. *Angew. Chem. Int. Ed.* **2015**, 54, 11162 –11167.

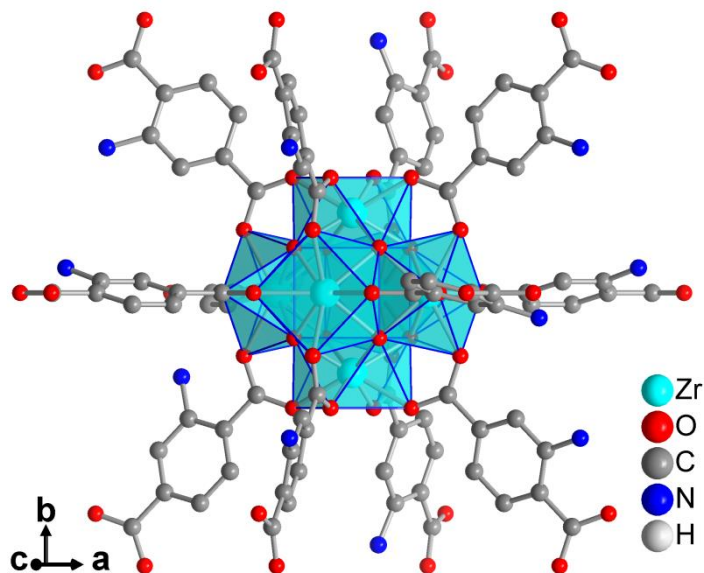
Linker disorder and defects in $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or **UiO-66-NH₂**

ideal, non-disordered presentation:

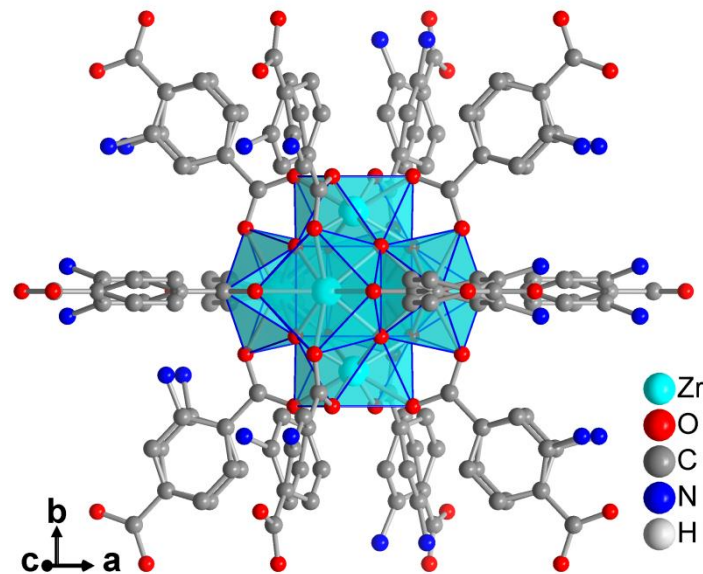


Linker disorder and defects in $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or **UiO-66-NH₂**

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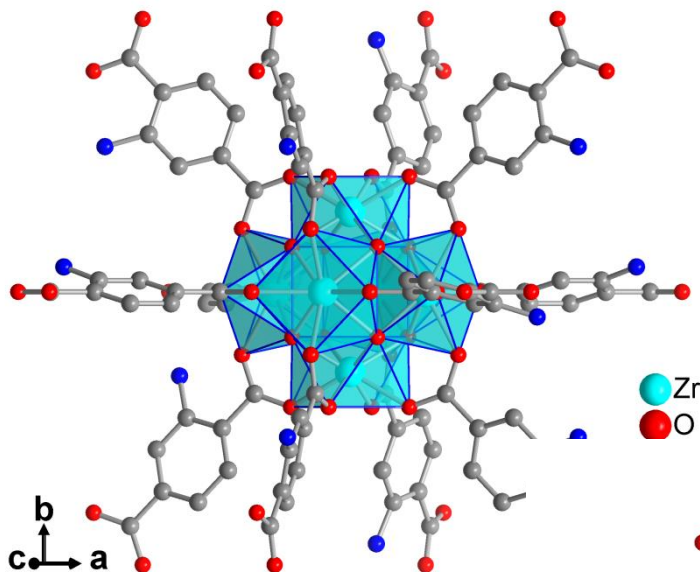


phenyl plane tilted by 17.2° to mirror plane:

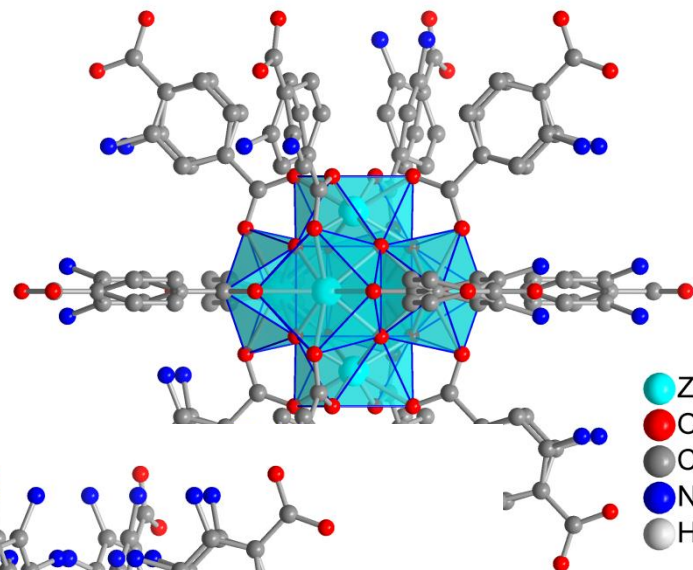


Linker disorder and defects in $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or **UiO-66-NH₂**

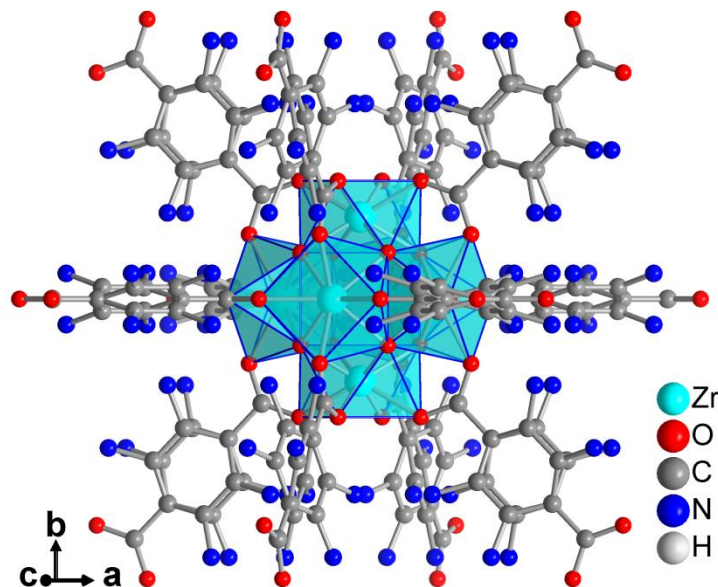
ideal, non-disordered presentation:



phenyl plane tilted by 17.2° to mirror plane:



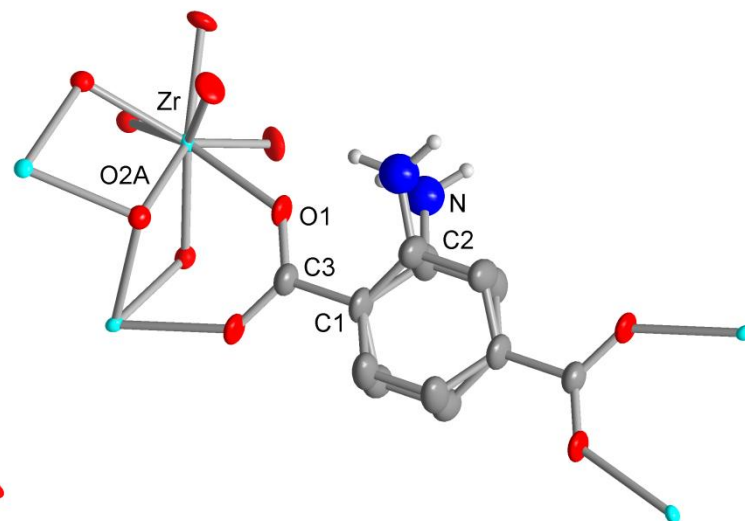
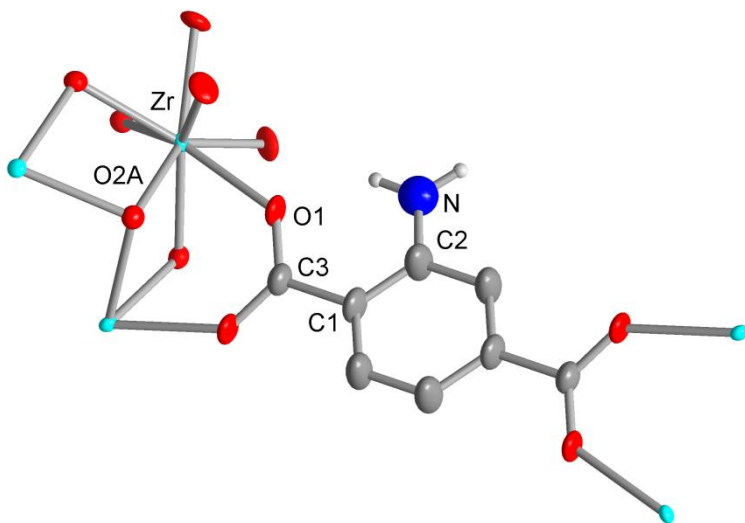
amino group distributed
over all 8 positions
(ideal occ. 0.125):



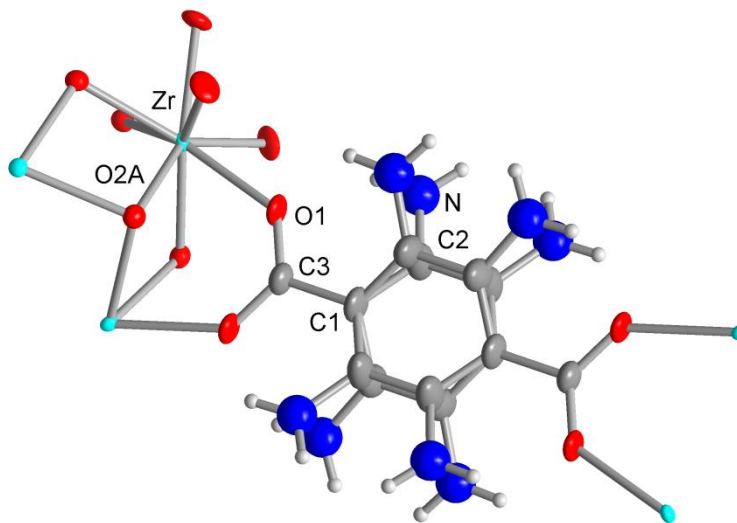
Linker disorder and defects in $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or UiO-66-NH_2

ideal, non-disordered presentation:

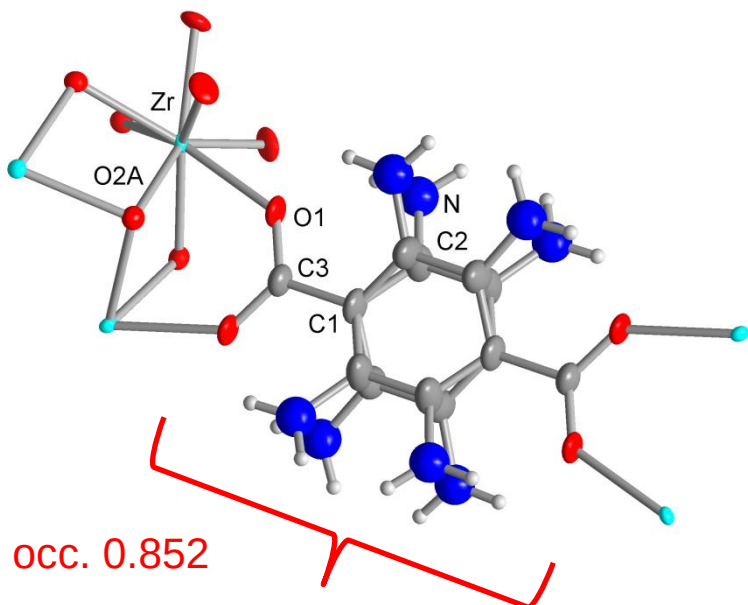
phenyl plane tilted by 17.2° to mirror plane:



amino group distributed
over all 8 positions
(ideal occ. 0.125):
- thus, N atom only
refined isotropically



Linker disorder and defects in $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or **UiO-66-NH₂**



⇒ 15% of linker missing!

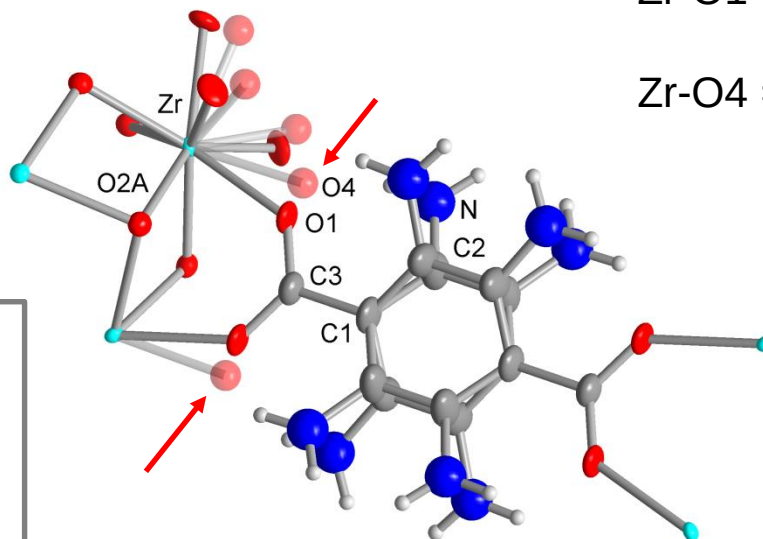
UiO-66-type MOFs are known for its defects.

Occupancy of all atoms except for Zr was refined as a free variable.

Occupancy of the 2-amino-carboxylate group (bdc-NH_2^{2-}) atoms refined to 0.852 of the ideal occupancy "1".

$\text{Zr-O1} = 2.232(3) \text{ \AA}$

$\text{Zr-O4} = 2.19(2) \text{ \AA}$



⇒ Coordination at Zr is then filled by the atom O4 whose reciprocal occupancy was 0.148.
O4 = H_2O or OH^-

Linker disorder and defects in
 $3\text{D}[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{bdc}+\text{NH}_2)_6]$ or UiO-66-NH_2

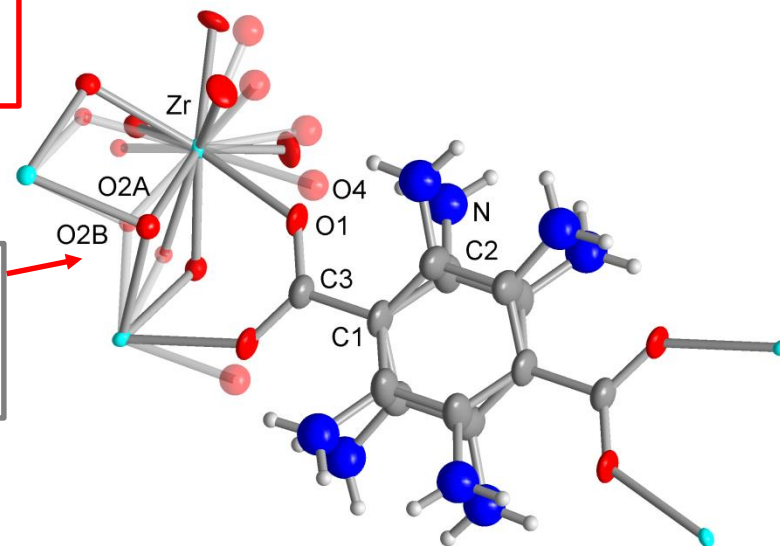
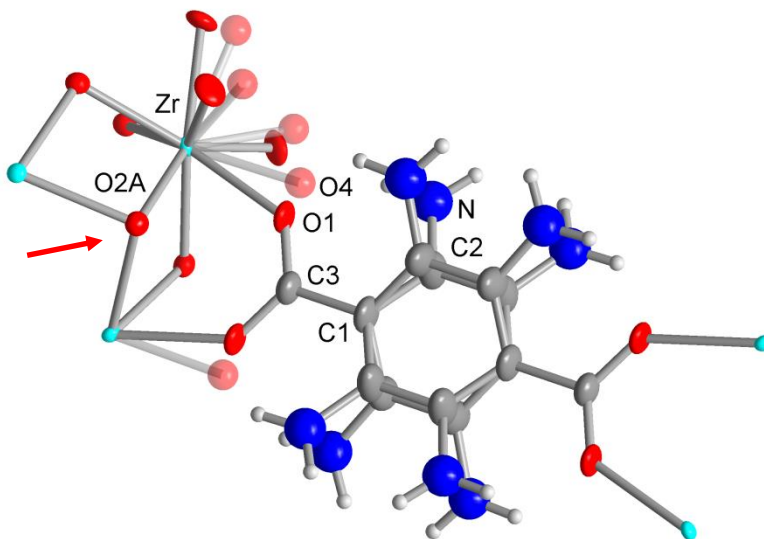
O2 stands for
 $\mu_3\text{-O}^{2-}$ and $\mu_3\text{-OH}^-$

Occupancy of O2A
refined to 0.53 of the
ideal occupancy
0.16667.

$\text{Zr-O2A} = 2.251(9) \text{ \AA} \rightarrow \mu_3\text{-OH}^-$

$\text{Zr-O2B} = 2.057(4) \text{ \AA} \rightarrow \mu_3\text{-O}^{2-}$

The reciprocal
occupancy of O2B was
0.47 (of ideal 0.16667)

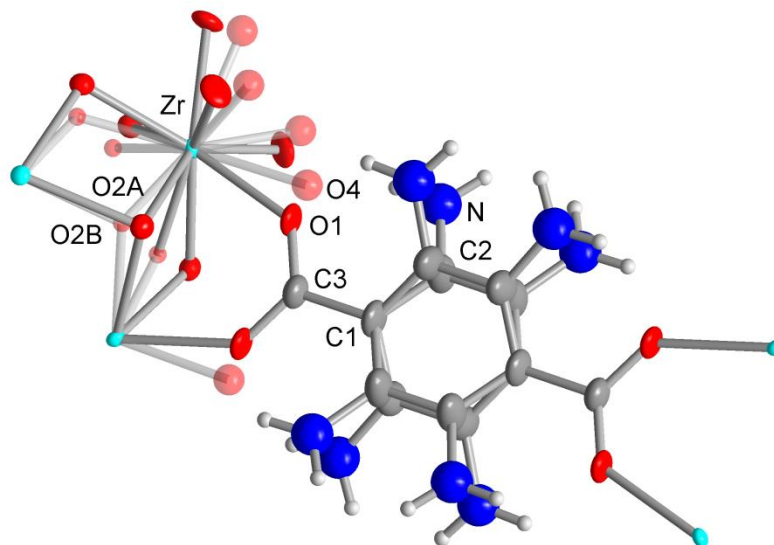


Linker disorder and defects in 3D-[Zr₆O₄(OH)₄(bdc+NH₂)₆] or UiO-66-NH₂

O2 stands for
 μ_3 -O and μ_3 -OH

Occupancy of **O2A**
refined to **0.53** of the
ideal occupancy.

The reciprocal
occupancy of **O2B** was
0.47 (of ideal 0.16667)



charge balance:

$$6 \times \text{Zr}^{4+} = 24+$$

$$(\text{O2A}) \ 8 \times 0.53 = 4.2 \times \text{OH}^- = 4.2-$$

$$(\text{O2B}) \ 8 \times 0.47 = 3.8 \times \text{O}^{2-} = 7.6-$$

$$\underline{6 \times 0.852 = 5.1 \times \text{bdc-NH}_2^{2-} = 10.2-}$$

$$\text{sum} = +2$$

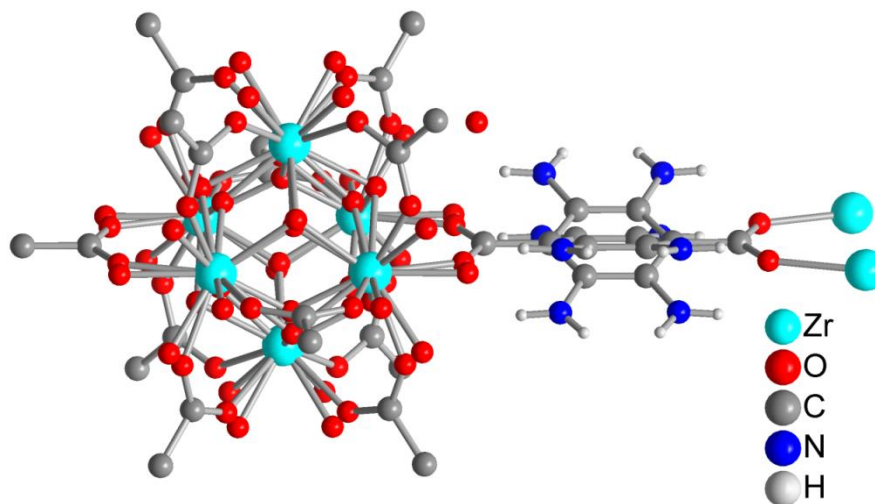
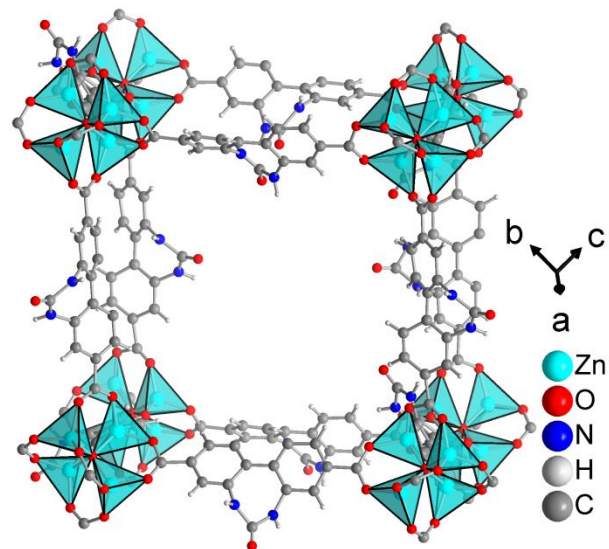
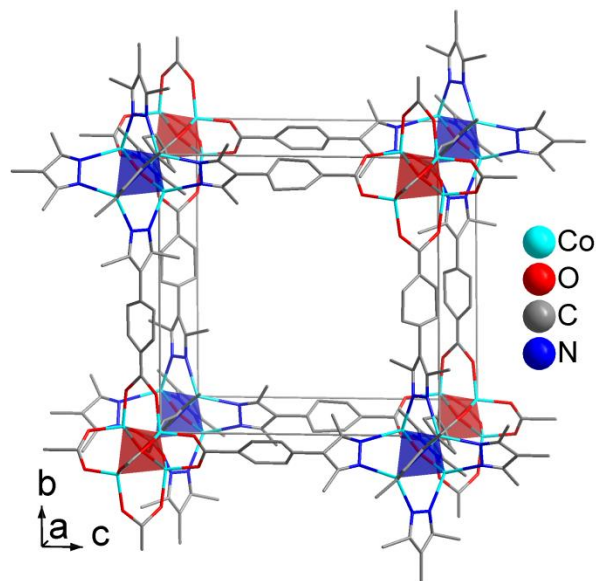
⇒

3.6 O4 = OH⁻ or H₂O ligand would
be **2 x OH⁻** and **1.6 H₂O**
for charge balance, electroneutrality

⇒ molecular "defect" formula in crystal:

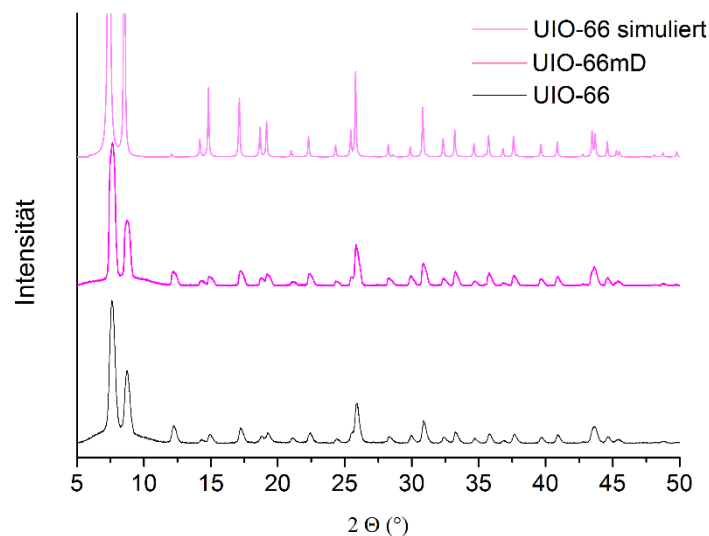
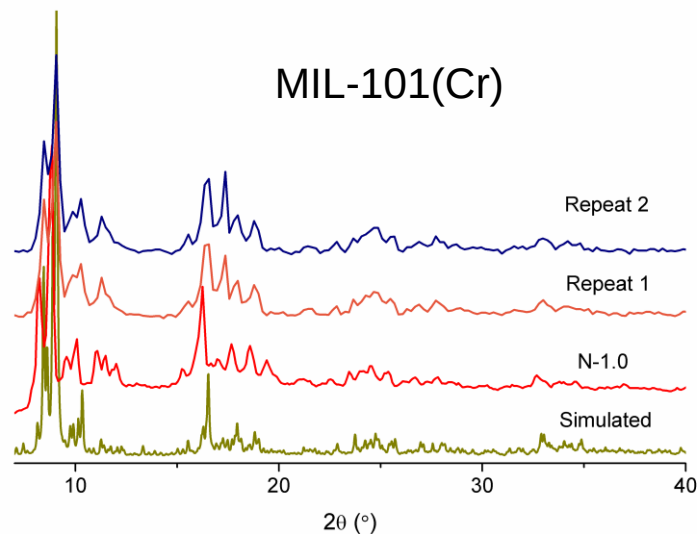


Our experience with MOF structures - 3 case studies



MOF synthesis and analysis

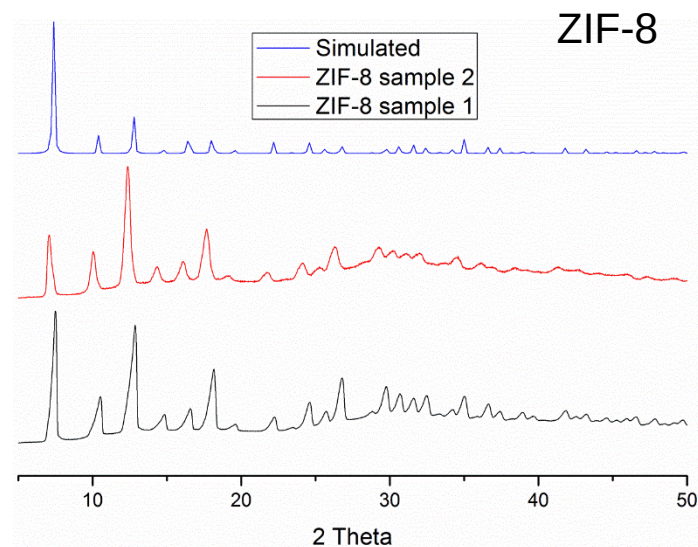
- validation of phase purity by powder X-ray diffraction



Dalton Trans. **2015**, 44, 16791-16801.
DOI: 10.1039/C5DT02625C

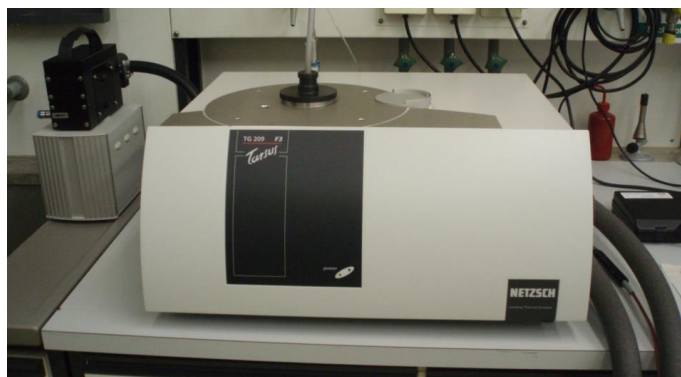


Bruker D2 Phaser PXRD

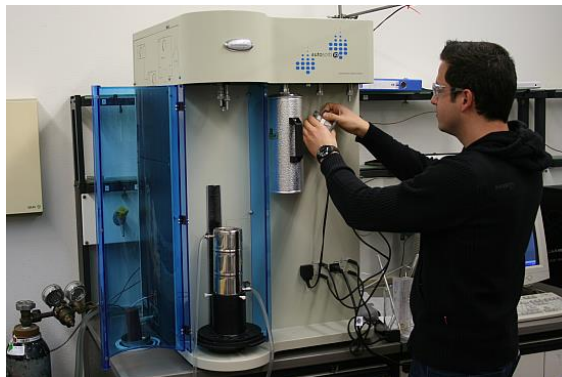


MOF synthesis and analysis

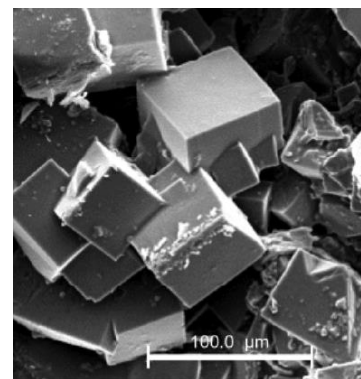
- TGA, gas sorption, SEM ...



Thermogravimetry (TGA)



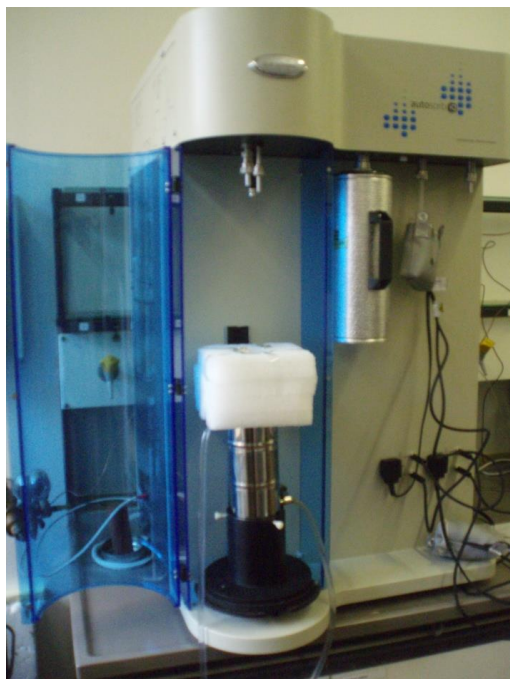
sorption measurements



SEM

MOF synthesis and analysis - gas sorption

Quantachrome Autosorb iQ
MP, all gas option (N₂-BET, H₂,
Ar, Kr, CO₂, CH₄, H₂O)



Nova 4200e



Micromeritics ASAP 2020; all
gas option (N₂-BET, H₂, Ar, Kr,
CO₂, CH₄, H₂O)



MOFs analysis:

- surface area determinations

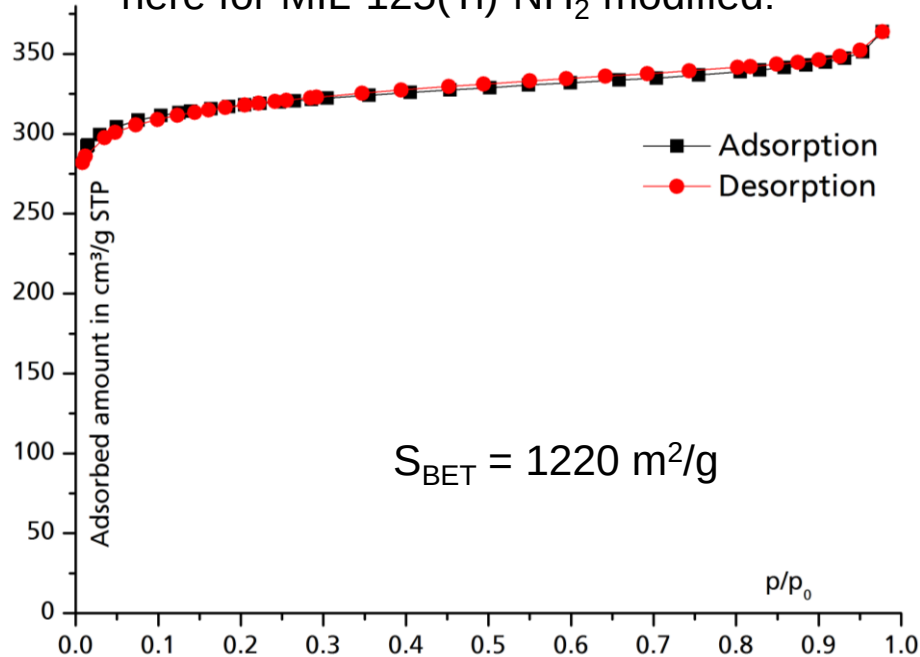
N₂-sorption measurements at 77 K (near condensation point of N₂, because of weak physisorption energy)

gives N₂-adsorption (and desorption) isotherm:

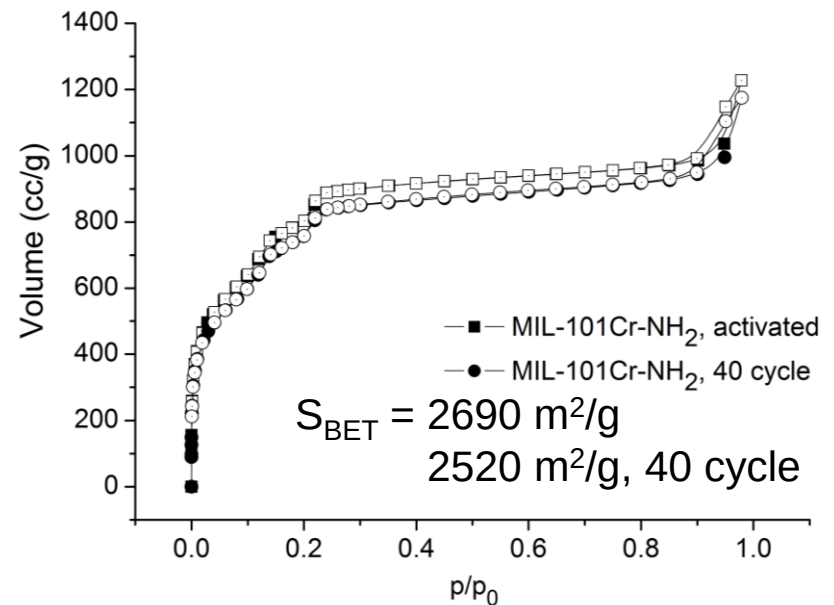
- evaluated according to Brunauer-Emmett-Teller (BET) theory

=> BET surface area, (micro)pore volume, NL-DFT pore size distribution

here for MIL-125(Ti)-NH₂ modified:



for MIL-101(Cr)-NH₂ modified:



Content

1. Definition and stability issue
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected potential applications

- new file